Lung Cancer Risk Factors in US Adults

An integrated analysis of NHIS 2019–2023 with regional air-quality, smoking and cancer-registry data

Revised version after midterm feedback

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Abstract

We set out to find which personal and regional factors go together with lung cancer in US adults, with an answer we could defend line by line. We stacked five years of the National Health Interview Survey (NHIS 2019–2023; 149,928 adults after cleaning) and joined three regional sources: EPA fine-particle pollution (PM2.5), CDC adult smoking prevalence (BRFSS) and official lung-cancer incidence (US Cancer Statistics). Lung cancer is rare in these data – 619 adults, 0.41% of the sample and 0.31% after survey weighting – and that rarity shaped almost every design decision, starting with rejecting accuracy as a metric (an "always no" rule would already be 99.59% accurate). A detailed profile of the study population (Section 2.5) shows who the cases are: 76% are aged 65 or older, 82% have ever smoked and 47% have COPD, against 31%, 38% and 5% of the other adults.

Age, smoking history and COPD stand out clearly. Ever-smokers have 7.7 times the odds of never-smokers (95% CI 6.2–9.4), and still 5.8 times after adjusting for age group; people with lung cancer are much older (median 72 vs 54 years; a random case is older than a random non-case 79.5% of the time). COPD carries an odds ratio of 15.1; among adults with known smoking status the crude 14.8 falls to 9.3 after adjusting for smoking, and among those whose age is also known it falls to 5.7 after adjusting for smoking and age group together. Lower education and income, underweight and poor self-rated health also go with higher rates, the last two most likely through reverse causation. Of 18 hypothesis tests, 14 remain significant after Holm correction, and a survey-weighted, cluster-robust check leaves every key odds ratio far from 1. At the regional level, the survey rate agrees with the official registry rate (Spearman $\rho = 0.79$ over 20 region-years), but this agreement comes from differences between the four regions rather than from year-to-year changes, so it is consistent with – not proof of – a valid target; PM2.5 and regional smoking prevalence show no detectable association, and as features the three regional columns carry a single shared signal (which region-year a person lives in) rather than three separate ones.

The result is a reproducible pipeline that one command rebuilds in under two minutes, with every cleaning step logged, all learned steps fitted on training rows only (checked by an automated assertion), and a leakage-free set of 38 features selected from 67 candidates. Feature selection also removes dependencies that span several columns – the comorbidity count, which is the exact sum of five condition flags, one reference category per one-hot group, and the region dummies – so that no feature has a variance inflation factor above 7.5 and a linear model can be fitted without perfect multicollinearity. A principal component analysis confirmed that the information is spread over many dimensions (32 of 38 components for 95% of the variance, and no near-zero component), so we keep the original, interpretable features for the modelling phase.

Keywords: lung cancer, risk factors, NHIS, CRISP-DM, data integration, hypothesis testing, confounding, class imbalance, data leakage, PCA

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1. Introduction

Our goal is simple to state: find out which personal and regional factors are associated with lung cancer, and hand the modelling phase a dataset it can trust.

1.1 Context

Lung cancer is the leading cause of cancer death in the United States, mostly because it is usually found late. Low-dose CT screening saves lives, but it is only offered to people at high risk. The US Preventive Services Task Force [6] defines them as adults aged 50–80 with at least 20 pack-years who still smoke or quit within the last 15 years. Validated risk models such as PLCOm2012 [7] go further and combine age, smoking dose and duration, COPD and family history. The real problem is therefore upstream of the scan: knowing who is at risk. That is a data problem, and it is the one we picked. Our target records a past diagnosis, so this midterm studies risk factors and associations; the repository keeps its original name (Lung-cancer-early-detection), but no claim about early detection is made here.

1.2 Problem statement and objectives

Using five years of nationally representative survey data, joined with regional pollution, smoking and cancer-registry data, we want to:

1. identify which demographic, behavioural, health and regional factors are associated with having ever been diagnosed with lung cancer;
2. measure how strong each association is, with effect sizes and confidence intervals rather than p-values alone;
3. check whether the strongest associations survive adjustment for the obvious confounders (age for smoking; smoking and age for COPD);
4. deliver a clean, documented, leakage-free feature set for a future risk model.

1.3 Research questions

Outcome. Having ever been told by a doctor that one had lung cancer, self-reported by US adults in NHIS 2019–2023 (149,928 adults, 619 cases).

We fixed these questions (Table 1) at the start of the analysis, so it could not drift toward whatever happened to look significant, and made each one name the comparison, the measure and the adjustment it relies on. The second question is split into three (RQ2a–c), one per family of factors. The same five labels mark every test in the code (`reports/tables/all_tests_holm.csv`): RQ1 has 1 test, RQ2a 1, RQ2b 7, RQ2c 6 and RQ3 3, for 18 in total.

1.4 Measurable success criteria

We wrote these the way we would write acceptance criteria for software: each one is a pass/fail check fixed before the analysis (Table 2). All eight are verified by the code: criteria 1–4 and 6–8 in `eda_results.txt` (section 10) and `reports/tables/success_criteria_check.csv`, and criterion 5 by an assertion in `supplementary_analysis.py` (section 11), which refits the scaler on training rows and fails the run if the saved parameters differ.

Two further rules apply to the modelling phase and are carried into Section 7.1: the pipeline must rebuild everything from raw files in under five minutes on a fresh clone, and the future model will be judged on recall, precision and PR-AUC, never on accuracy, against the benchmark set by the USPSTF-style rule (recall 63%, precision 1.2%; Table 41).

Table 1. Research questions and how each one is answered.

#	Question	How we answer it	Where
RQ1	**Smoking.** How much higher are the odds of a lung-cancer history among ever-smokers (current or former) than among never-smokers? How much of that odds ratio remains when ever- and never-smokers are compared within the same age group (Mantel–Haenszel over five age groups), and does its size differ between age groups (Breslow–Day)?	χ^2 test of independence, Cramér's V; odds ratio with 95% CI; Mantel–Haenszel odds ratio over five age groups with CMH test; Breslow–Day test	§4.2–4.3, Table 22
RQ2a	**Age.** Are adults with a lung-cancer history older than those without, and by how much (Mann–Whitney U, rank-biserial r)? How does the survey-weighted rate change across the five age groups, alone and combined with smoking status?	Mann–Whitney U with rank-biserial r (Welch t-test as a check); survey-weighted rates by age group and by age group × smoking status	§2.5.1, §4.3, Fig. 7–8
RQ2b	**Chronic conditions and e-cigarettes.** Which of five doctor-diagnosed conditions (COPD, heart disease, hypertension, diabetes, asthma), and ever using an e-cigarette, are associated with a lung-cancer history (odds ratios with 95% CIs)? Does the COPD association remain after stratifying by smoking status, and then by smoking status × age group?	χ^2 tests with odds ratios (95% CI) and Cramér's V; Cochran–Mantel–Haenszel and Breslow–Day tests across smoking groups; Mantel–Haenszel odds ratio over smoking status × age group (15 strata)	§4.4–4.5, Fig. 9–10, Tables 23–25
RQ2c	**Socio-demographic and health factors.** Is a lung-cancer history associated with sex, race/ethnicity, education, family income relative to the poverty line, BMI category and self-rated general health (χ^2 tests of independence, Cramér's V)? For the four ordered factors, how does the survey-weighted rate change from the lowest to the highest level (described from the rates, not tested as a trend)?	χ^2 tests of independence with Cramér's V (smallest race groups merged); survey-weighted rates per level, compared from lowest to highest for the four ordered factors (education, income, BMI, self-rated health) – described, not tested as a trend	§4.6, Fig. 11, Table 26
RQ3	**Regional context.** Across the 20 region-years (4 US Census regions × 2019–2023), does the survey-weighted NHIS lung-cancer rate follow (a) the official US Cancer Statistics incidence, (b) mean EPA PM2.5, and (c) BRFSS adult smoking prevalence? Is any agreement driven by differences between regions or by year-to-year changes within them? As features, how much information do the three regional columns carry about an individual's lung-cancer history compared with personal factors (mutual information against a permutation null)?	Spearman correlation over 20 region-years, split into between-region (4 regional means) and within-region parts; mutual information on training rows with a permutation null	§4.8, §5.5, Fig. 13–14, Tables 29 and 36

Table 2. Success criteria fixed before the analysis, with their pass threshold.

#	Criterion	Target
1	No duplicate person IDs in the clean dataset	0 duplicates
2	No missing values in the model-ready dataset	0 missing cells
3	The regional join neither adds nor loses rows	row count identical before and after
4	Train and test keep the same lung-cancer rate (stratified split)	difference < 0.05 percentage points
5	No test-set information is used for imputation, selection, scaling or PCA	all fitted on training rows (asserted in code)
6	Every research question (RQ1, RQ2a, RQ2b, RQ2c, RQ3) has at least one justified test, with assumptions checked, an effect size and a Holm-corrected p-value	$\alpha = 0.05$ after Holm correction
7	Every candidate feature has a written KEEP / DROP reason	0 missing reasons
8	No leakage feature: "ever had any cancer" (CANEV_A) is never a feature	absent from every feature set

1.5 Method and scope

We follow CRISP-DM [10] from business understanding to data preparation, which is the standard process taught in the course [19, Chapter 1]. In scope: data gathering and integration, data understanding, preprocessing, exploratory data analysis, hypothesis testing and feature engineering. Out of scope for the midterm, as the rubric specifies: training or evaluating predictive models, clinical interpretation and causal claims. Modelling is described as planned future work only; the stratified (Mantel–Haenszel) and survey-weighted odds ratios in Section 4 are descriptive adjustments of associations, not predictive models.

1.6 Organisation of the paper

The remainder of the paper follows the CRISP-DM phases. Section 2 describes the data sources, the study population in detail and the data-quality issues; Section 3 presents the preprocessing methodology; Section 4 reports the exploratory analysis and the hypothesis tests; Section 5 describes feature engineering; Section 6 discusses the findings, limitations and ethical considerations; and Section 7 concludes with the planned modelling phase.

2. Data Description and Sources

Our final dataset has 149,928 adults and 26 columns (Table 7). It is built from four public US government sources: one person-level survey stacked over five years, and three regional tables joined on region and year.

Dataset composition at a glance

Every rate, odds ratio and correlation in this report is computed on a stated subset of the adults below, so we first show what the data are made of (Tables 3–5; source: `reports/tables/dataset_composition.csv`, produced by `supplementary_analysis.py`). Each survey year contributes between 27,594 and 31,925 adults and between 112 and 145 cases, so no single year dominates (Table 3). The 26 columns of the clean file are of nine kinds (Table 4): one identifier, one survey weight and the target, which are never features; two numeric columns, four ordinal, three nominal and seven binary survey variables; four engineered columns; and three regional columns joined on region and year. The model-ready file keeps 38 of the 67 candidate features (Section 5.4), grouped in Table 5. Finally, the stratified 80/20 split keeps the same 0.41% case rate in the 119,942 training rows and the 29,986

test rows (Table 3, lower part). Unless stated otherwise, descriptive numbers use all 149,928 adults with the variable known, and anything that ranks or fits features uses the training rows only.

Table 3. Adults and lung-cancer cases per survey year, and in the train / test split (unweighted).

	Adults	Cases	Rate (sample)
2019	31,925	112	0.35%
2020	31,513	131	0.42%
2021	29,430	115	0.39%
2022	27,594	116	0.42%
2023	29,466	145	0.49%
All years	**149,928**	**619**	**0.41%**
Training rows	119,942	495	0.41%
Test rows	29,986	124	0.41%

Table 4. The 26 columns of the clean dataset by type.

Type	Columns	Which
Identifier	1	person_id
Survey weight	1	survey_weight
Target	1	lung_cancer
Numeric	2	age, year
Ordinal	4	education, income_group, general_health, bmi_category
Nominal	3	race_ethnicity, region, smoking_status
Binary	7	sex, ever_ecigarette, copd, asthma, hypertension, heart_disease, diabetes
Engineered	4	age_group, ever_smoker, screening_age_smoker, comorbidity_count
Regional (joined)	3	pm25_mean, smoking_rate, lung_cancer_incidence
Total	**26**	

2.1 Sources and provenance

All four sources are public-domain US government data (Table 6), so we can commit them to the repository. The raw files were downloaded on 24–25 October 2026 with `scripts/download_data.py`, which fetches them again from the official URLs in the References if they are missing. They sit unchanged in `data/raw/`, and we only ever change data through code, never by hand – the "never overwrite the original" rule of the course (Chapter 2). The three regional files were reduced to the rows we need at download time so that every file stays under GitHub's 100 MB limit; the reduction is done in code and is fully reproducible.

2.2 Granularity and target definition

One row is one surveyed adult (18+) in one survey year. NHIS interviews one randomly selected adult per household, so the household number plus the survey year (`person_id`) identifies a row. The regional sources are coarser – a monitor, or a state – and the public NHIS files only reveal the person's US Census region (Northeast, Midwest, South, West). We therefore aggregated every regional measure to region × year, so each regional column takes only $4 \times 5 = 20$ distinct values and describes the person's environment, not the person.

The target. `lung_cancer` is 1 if the person reports that a doctor ever told them they had lung cancer. NHIS does not ask this in a single question, so we derive it from two: "ever told you had cancer?" (`CANEV_A`) and, only for people who said yes, "what kind?" (`LUNGCAN_A`). The target is 0 for people who never had cancer and for people who had a different cancer.

Table 5. The 38 final model features by group (full list in Appendix B).

Group	Features	Count
Demographic	age (scaled); sex: Female (reference: Male); race/ethnicity: AIAN, AIAN and other, Asian, Black, Hispanic, Other or multiple (reference: White)	8
Socio-economic	education: Bachelor, Graduate, Less than high school, Some college, Unknown (reference: High school / GED); income vs poverty line: Below poverty, Low, Middle (reference: High)	8
Smoking and nicotine	screening_age_smoker; screening_age_smoker_unknown; ever_ecigarette; ever_ecigarette_unknown; smoking status: Current, Former (reference: Never)	6
Health	copd; asthma; hypertension; heart_disease; diabetes; general health: Fair, Good, Poor, Very good (reference: Excellent); BMI category: Obese, Overweight, Underweight, Unknown (reference: Healthy)	13
Regional context	pm25_mean (scaled); smoking_rate (scaled); lung_cancer_incidence (scaled)	3
Total		**38**

Table 6. Data sources, publisher, granularity and raw size.

#	Source	Publisher	One raw row is	Raw size	Used for
1	National Health Interview Survey, Sample Adult files 2019–2023 [1]	CDC – National Center for Health Statistics	one adult in one survey year	150,220 adults (27,651–31,997 per year)	individual features and the target
2	EPA AirData, annual concentration by monitor, 2019–2023 [2]	US Environmental Protection Agency	one monitor × pollutant × year	75,534–107,019 rows per year	regional PM2.5 pollution
3	BRFSS prevalence data, current smokers, 2019–2023 [3]	CDC Behavioral Risk Factor Surveillance System	one state × year × answer	1,644 rows	regional adult smoking rate
4	US Cancer Statistics, incidence by area [4]	CDC and National Cancer Institute	one area × year × sex × race × site	23,796 lung-cancer rows	regional lung-cancer incidence

2.3 Data dictionary

2.4 Target distribution

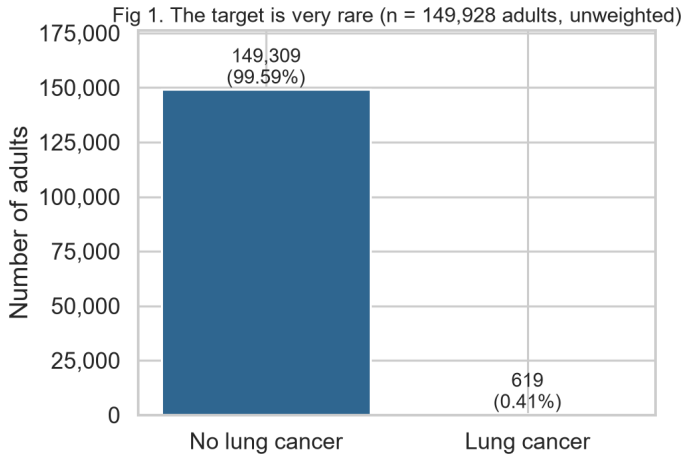


Figure 1. Class balance of the target. Only 619 of 149,928 adults (0.41%) report a lung-cancer diagnosis; the population rate after NHIS survey weighting is 0.31%.

The imbalance is extreme (Fig. 1): a model that always answers "no" would be 99.59% accurate while finding no one. This single number drove our choice of effect sizes (odds ratios rather than accuracy-like measures), of a stratified train/test split, and of the metrics planned for the modelling phase.

2.5 Who is in the data: a detailed profile of the study population

Before testing anything, we describe exactly who the 149,928 adults are and where the 619 cases sit. Tables 8–12 give, for every category of every variable, the number of adults, their share of the sample and of the US adult population (survey-weighted), the number of adults with and without a lung-cancer history, the lung-cancer rate in the sample and in the population, and the mean age of the group. "Unknown" rows are people who did not give a usable answer; we keep them so that every block adds up to 149,928 adults and 619 cases (shares therefore differ slightly from Fig. 5, which uses known answers only). All numbers come from `reports/tables/population_profile.csv`, produced by `scripts/supplementary_analysis.py`.

Two reading rules apply throughout. First, the sample rate is what we observe among the people interviewed, while the weighted rate estimates the rate among all US adults: NHIS interviews some groups more than others (for example, 31.1% of the adults with a known age are 65 or older, against 21.9% of the population), and the survey weights correct for this. Second, raw counts of cases depend on group size: 507 of the 619 cases are White adults simply because White adults are 67.5% of the sample, so groups are compared through their rates, never through their counts.

2.5.1 Demographics: sex, age and race/ethnicity

- **Sex.** The sample has 81,360 women (54.3%) and 68,552 men (45.7%); after weighting the split is 51.5% / 48.5%. More cases are women (354, against 265 men), but women are also more numerous and slightly older (mean age 53.6 vs 51.9 years), and the weighted rates are almost equal (0.32% vs 0.30%). Section 4.6 confirms that sex is not associated with lung cancer.
- **Age.** Adults aged 18–44 are the largest group (36.3% of the sample, 45.8% of the population) but include only 13 cases (0.02%). From there the rate climbs steadily: 0.10% at 45–54, 0.42% at 55–64, 0.84% at 65–74 and 1.24% at 75+. The 46,500 adults aged 65 or older (31.1% of those with a known age) hold 471 of the 619 cases (76.1%; 76.2% of the 618 cases with a known age, as in Table 13).

Table 7. Data dictionary of the final clean dataset (26 columns); the type follows Table 4. Full version: `data/clean/data_dictionary.csv`.

Column	Meaning	Type	Built from
person_id	Unique ID: survey year + NHIS household number	identifier	NHIS HHX + SRVY_YR
year	Survey year (2019-2023)	numeric	SRVY_YR
survey_weight	NHIS sampling weight (use for population estimates)	weight	WTFA_A
age	Age in years (85 = 85 or older)	numeric	AGEP_A
sex	Male / Female	binary	SEX_A
race_ethnicity	Hispanic origin and race	nominal	HISPALLP_A
region	US region	nominal	REGION
education	Highest education level (grouped)	ordinal	EDUC_A (EDUCP_A from 2021)
income_group	Family income vs poverty line: below (<1), low (1-2), middle (2-4), high (4+)	ordinal	RATCAT_A
general_health	Self-rated general health	ordinal	PHSTAT_A
bmi_category	Body mass index category	ordinal	BMICAT_A
smoking_status	Cigarette smoking: Current / Former / Never	nominal	SMKCIGST_A
ever_ecigarette	Ever used an e-cigarette (1 = yes)	binary	ECIGEV_A
copd	Ever told COPD, emphysema or chronic bronchitis (1 = yes)	binary	COPDEV_A
asthma	Ever had asthma (1 = yes)	binary	ASEV_A
hypertension	Ever told high blood pressure (1 = yes)	binary	HYPEV_A
heart_disease	Ever told coronary heart disease (1 = yes)	binary	CHDEV_A
diabetes	Ever told diabetes (1 = yes)	binary	DIBEV_A
age_group	NEW: age band	engineered	from age
ever_smoker	NEW: current or former smoker (1 = yes)	engineered	from smoking_status
screening_age_smoker	NEW: age 50-80 and ever smoked (proxy of screening eligibility)	engineered	from age + smoking
comorbidity_count	NEW: number of the 5 conditions above	engineered	from copd...diabetes
pm25_mean	JOINED: average fine-particle air pollution (PM2.5, µg/m^3) in the person's region that year	regional	EPA AirData, by region + year
smoking_rate	JOINED: % of adults who currently smoke in the person's region that year	regional	CDC BRFSS, by region + year
lung_cancer_incidence	JOINED: new lung cancer cases per 100,000 people in the person's region that year	regional	CDC US Cancer Statistics, by region + year
lung_cancer	TARGET: ever told they had lung cancer (1 = yes)	target	CANEV_A + LUNGCAN_A

Table 8. Demographic profile of the 149,928 adults: adults, share of the sample and of the US adult population (weighted, wtd), adults with and without a lung-cancer history, lung-cancer rate and mean age, by sex, age group and race/ethnicity. AIAN = American Indian or Alaska Native.

Group	Adults	% of sample	% of US adults (wtd)	Lung cancer	No lung cancer	Rate (sample)	Rate (wtd)	Mean age
Sex								
Female	81,360	54.3	51.5	354	81,006	0.44%	0.32%	53.6
Male	68,552	45.7	48.5	265	68,287	0.39%	0.30%	51.9
Unknown	16	0.0	0.0	0	16	0.00%	0.00%	37.6
Age group (years)								
18–44	54,481	36.3	45.8	13	54,468	0.02%	0.02%	32.4
45–54	21,574	14.4	15.7	21	21,553	0.10%	0.08%	49.6
55–64	27,020	18.0	16.5	113	26,907	0.42%	0.45%	59.7
65–74	26,445	17.6	13.0	222	26,223	0.84%	0.81%	69.3
75+	20,055	13.4	8.9	249	19,806	1.24%	1.25%	80.5
Unknown	353	0.2	0.2	1	352	0.28%	0.11%	–
Race / ethnicity								
White	101,181	67.5	62.5	507	100,674	0.50%	0.39%	55.3
Hispanic	20,411	13.6	17.0	27	20,384	0.13%	0.10%	45.0
Black	16,138	10.8	11.8	55	16,083	0.34%	0.22%	51.3
Asian	8,453	5.6	6.0	16	8,437	0.19%	0.22%	46.9
AIAN	962	0.6	0.7	8	954	0.83%	0.60%	50.5
AIAN and other	1,114	0.7	0.7	5	1,109	0.45%	0.43%	52.2
Other or multiple	1,669	1.1	1.3	1	1,668	0.06%	0.02%	41.2

- **Race and ethnicity.** NHIS records Hispanic origin and race in one variable; it is the only measure of cultural background in our variable set. The sample has 101,181 White adults (507 with and 100,674 without a lung-cancer history), 20,411 Hispanic adults (27 with, 20,384 without), 16,138 Black adults (55 with, 16,083 without), 8,453 Asian adults (16 with, 8,437 without), and three small groups – AIAN (American Indian or Alaska Native, 962 adults, 8 cases), AIAN and another group (1,114 adults, 5 cases) and other or multiple races (1,669 adults, 1 case). The weighted rate is highest among White adults (0.39%) and lowest among Hispanic adults (0.10%). The three small groups total only 14 cases, so their rates are unstable and they are merged for testing (§4.6).

Why the groups differ. Table 9 looks inside each sex and race/ethnicity group. White adults are the oldest group (mean age 55.3, 36.5% aged 65+) and the most likely, after the AIAN groups, to have ever smoked (42.7%); Hispanic adults are among the youngest (mean 45.0, 15.5% aged 65+) and smoke much less (25.0% ever-smokers). Asian adults have the lowest smoking prevalence (18.3%). Age and smoking – the two strongest risk factors in this project – therefore explain a large part of the raw differences between race/ethnicity groups, which is why we do not interpret those differences as an effect of race itself. Men smoke more than women (43.4% vs 32.9% ever-smokers), yet women are older on average and report COPD slightly more often (6.2% vs 5.0%), which is consistent with the near-equal rates by sex.

2.5.2 Socio-economic position, region and survey year

- **Education.** The largest groups are "some college" (42,452 adults, 28.3%) and high school / GED (37,368, 24.9%; Table 10). The rate falls from 0.67% (sample) among adults without a high-school diploma to 0.19% among bachelor's graduates; weighted, from 0.49% to 0.15%.
- **Income.** 42.9% of adults have a family income at least four times the poverty line and 10.1% live below it. The rate halves from below-poverty (0.46% weighted) to high income (0.21%).
- **Region.** The South is the largest region (54,341 adults, 36.2%), followed by the West (25.1%),

Table 9. Inside each sex and race/ethnicity group: age, sex, smoking and COPD, which explain much of the difference in lung-cancer rates (percentages among adults with a known answer; rate weighted). Source: `group_profile.csv`.

Group	Adults	Cases	Mean age	Median age	% aged 65+	% women	% ever smoked	% COPD	Rate (wtd)
Sex									
Female	81,360	354	53.6	55	32.9	–	32.9	6.2	0.32%
Male	68,552	265	51.9	53	29.0	–	43.4	5.0	0.30%
Race / ethnicity									
White	101,181	507	55.3	58	36.5	53.5	42.7	6.7	0.39%
Hispanic	20,411	27	45.0	43	15.5	54.3	25.0	2.4	0.10%
Black	16,138	55	51.3	52	26.2	59.0	30.9	5.2	0.22%
Asian	8,453	16	46.9	44	18.5	54.2	18.3	1.0	0.22%
AIAN	962	8	50.5	51	23.8	57.1	50.4	10.5	0.60%
AIAN and other	1,114	5	52.2	53.5	28.6	57.5	50.6	11.6	0.43%
Other or multiple	1,669	1	41.2	38	11.7	53.9	32.5	2.4	0.02%

Table 10. Socio-economic, geographic and yearly profile (same columns as Table 8). Income is family income divided by the poverty line.

Group	Adults	% of sample	% of US adults (wtd)	Lung cancer	No lung cancer	Rate (sample)	Rate (wtd)	Mean age
Education								
Less than high school	12,809	8.5	10.9	86	12,723	0.67%	0.49%	56.8
High school / GED	37,368	24.9	27.5	219	37,149	0.59%	0.40%	53.7
Some college	42,452	28.3	29.2	195	42,257	0.46%	0.32%	52.4
Bachelor	34,364	22.9	19.8	64	34,300	0.19%	0.15%	50.3
Graduate	22,175	14.8	11.9	48	22,127	0.22%	0.16%	53.5
Unknown	760	0.5	0.7	7	753	0.92%	0.59%	54.5
Family income vs poverty line								
Below poverty (<1)	15,119	10.1	10.2	95	15,024	0.63%	0.46%	50.5
Low (1–2)	26,307	17.5	18.0	164	26,143	0.62%	0.43%	54.2
Middle (2–4)	44,165	29.5	30.1	187	43,978	0.42%	0.32%	53.2
High (4+)	64,337	42.9	41.7	173	64,164	0.27%	0.21%	52.5
Census region								
Northeast	24,963	16.6	17.6	117	24,846	0.47%	0.37%	53.8
Midwest	33,049	22.0	20.8	174	32,875	0.53%	0.42%	53.2
South	54,341	36.2	38.0	229	54,112	0.42%	0.29%	52.9
West	37,575	25.1	23.6	99	37,476	0.26%	0.20%	51.7
Survey year								
2019	31,925	21.3	19.7	112	31,813	0.35%	0.28%	52.0
2020	31,513	21.0	19.9	131	31,382	0.42%	0.31%	53.5
2021	29,430	19.6	19.9	115	29,315	0.39%	0.28%	52.5
2022	27,594	18.4	20.1	116	27,478	0.42%	0.34%	52.9
2023	29,466	19.7	20.3	145	29,321	0.49%	0.34%	53.2

the Midwest (22.0%) and the Northeast (16.6%). The weighted rate is highest in the Midwest (0.42%) and lowest in the West (0.20%); the matching sample rates are 0.53% and 0.26%. The West is also the youngest region (mean age 51.7).

- **Survey year.** Each year contributes 27,594–31,925 adults and 112–145 cases. The rate drifts upward from 0.35% in 2019 to 0.49% in 2023 (weighted, 0.28% to 0.34%), but the difference between years is not statistically significant ($\chi^2(4) = 7.9$, $p = 0.096$; `rate_by_year.csv`). This matters because 2019 was the first year of the redesigned NHIS questionnaire and 2020 collection was disrupted by COVID-19 (largely telephone interviews); because the years do not differ significantly, pooling them is reasonable, and we keep year in the data so the modelling phase can check it again.

2.5.3 Smoking, body weight, self-rated health and chronic conditions

Table 11. Smoking, e-cigarette use, body weight and self-rated health (same columns as Table 8).

Group	Adults	% of sample	% of US adults (wtd)	Lung cancer	No lung cancer	Rate (sample)	Rate (wtd)	Mean age
Cigarette smoking								
Never	90,937	60.7	63.5	107	90,830	0.12%	0.08%	50.3
Former	37,452	25.0	22.0	367	37,085	0.98%	0.81%	59.7
Current	17,586	11.7	11.7	127	17,459	0.72%	0.60%	51.1
Unknown	3,953	2.6	2.8	18	3,935	0.46%	0.32%	52.9
Ever used an e-cigarette								
No	122,606	81.8	79.4	497	122,109	0.41%	0.31%	55.1
Yes	23,470	15.7	17.8	105	23,365	0.45%	0.29%	41.0
Unknown	3,852	2.6	2.8	17	3,835	0.44%	0.32%	52.7
BMI category								
Underweight	2,324	1.6	1.6	33	2,291	1.42%	1.14%	51.7
Healthy	46,153	30.8	30.7	221	45,932	0.48%	0.35%	51.6
Overweight	50,466	33.7	33.2	187	50,279	0.37%	0.29%	54.2
Obese	47,471	31.7	32.2	165	47,306	0.35%	0.24%	52.4
Unknown	3,514	2.3	2.3	13	3,501	0.37%	0.28%	54.6
Self-rated general health								
Excellent	32,824	21.9	23.4	18	32,806	0.05%	0.03%	46.5
Very good	51,131	34.1	33.9	73	51,058	0.14%	0.10%	51.1
Good	42,973	28.7	28.3	173	42,800	0.40%	0.28%	55.4
Fair	17,483	11.7	11.1	206	17,277	1.18%	0.96%	59.8
Poor	5,457	3.6	3.3	149	5,308	2.73%	2.47%	63.4
Unknown	60	0.0	0.0	0	60	0.00%	0.00%	61.5

- **Smoking.** 90,937 adults have never smoked (60.7% of all adults, 62.3% of those who answered), 37,452 are former and 17,586 current smokers; 3,953 did not answer. Former smokers have the highest rate (0.98% sample, 0.81% weighted) and are much older (mean age 59.7, against 51.1 for current and 50.3 for never-smokers), a first hint of the reverse causation discussed in §4.2.
- **E-cigarettes.** 23,470 adults (15.7%) have ever used one. They are much younger (mean age 41.0 vs 55.1), which is why their crude rate (0.29% weighted) is close to that of non-users (0.31%).
- **Body weight and self-rated health.** Underweight adults are only 1.6% of the sample but have the highest rate (1.42%, 33 cases). Self-rated health shows the steepest gradient in the whole dataset: 0.05% for "excellent" up to 2.73% for "poor" (sample rates), with 149 cases among the 5,457 adults in poor health.

Table 12. Chronic conditions (ever told by a doctor) and the number of the five conditions (same columns as Table 8).

Group	Adults	% of sample	% of US adults (wtd)	Lung cancer	No lung cancer	Rate (sample)	Rate (wtd)	Mean age
COPD								
No	141,386	94.3	95.3	330	141,056	0.23%	0.17%	52.0
Yes	8,440	5.6	4.6	288	8,152	3.41%	3.13%	65.4
Unknown	102	0.1	0.1	1	101	0.98%	0.32%	64.8
Heart disease								
No	140,524	93.7	95.0	475	140,049	0.34%	0.25%	51.6
Yes	9,073	6.1	4.8	139	8,934	1.53%	1.52%	70.7
Unknown	331	0.2	0.2	5	326	1.51%	1.34%	69.9
Hypertension								
No	94,990	63.4	68.2	215	94,775	0.23%	0.16%	46.7
Yes	54,776	36.5	31.7	404	54,372	0.74%	0.63%	63.3
Unknown	162	0.1	0.1	0	162	0.00%	0.00%	58.7
Diabetes								
No	133,796	89.2	90.4	488	133,308	0.36%	0.26%	51.4
Yes	16,064	10.7	9.5	131	15,933	0.82%	0.74%	64.5
Unknown	68	0.0	0.0	0	68	0.00%	0.00%	59.7
Asthma								
No	129,025	86.1	85.8	502	128,523	0.39%	0.29%	53.2
Yes	20,839	13.9	14.2	116	20,723	0.56%	0.41%	50.2
Unknown	64	0.0	0.0	1	63	1.56%	2.55%	57.4
Number of the 5 conditions								
0	76,217	50.8	55.2	87	76,130	0.11%	0.08%	45.9
1	47,541	31.7	29.9	203	47,338	0.43%	0.33%	57.0
2	18,719	12.5	10.8	166	18,553	0.89%	0.74%	64.2
3	5,808	3.9	3.2	118	5,690	2.03%	2.05%	67.1
4	1,426	1.0	0.8	36	1,390	2.52%	2.24%	68.0
5	217	0.1	0.1	9	208	4.15%	3.79%	66.7

- **Chronic conditions.** 8,440 adults (5.6%) report COPD, and 288 of them have a lung-cancer history – 46.5% of all cases, at a rate of 3.41%, fifteen times the 0.23% of adults without COPD. About half of adults (50.8%) have none of the five conditions; the rate then rises with every additional condition, from 0.11% with none to 4.15% with all five.

2.5.4 Adults with and without a lung-cancer history, side by side

Table 13 summarises the same information from the other direction: what the 619 cases look like compared with the 149,309 other adults. Cases are about 18.5 years older on average (71.2 vs 52.7), are more than twice as likely to have ever smoked (82.2% vs 37.5%), are eight times as likely to report COPD (46.6% vs 5.5%) and almost four times as likely to rate their health as fair or poor (57.4% vs 15.1%). They are more often White (81.9% vs 67.4%) and less often Hispanic (4.4% vs 13.7%) – a difference that the age and smoking profiles of Table 9 largely explain – and the share of women is similar (57.2% vs 54.3%).

Table 13. Characteristics of adults with and without a lung-cancer history (unweighted; percentages among adults with a known answer). Source: `cases_vs_noncases.csv`.

Characteristic	Lung cancer	No lung cancer
Adults	619	149,309
Age, mean (years)	71.2	52.7
Age, standard deviation (years)	10.2	18.3
Age, median (years)	72.0	54.0
Age, interquartile range (years)	65–79	37–68
Aged 65 or older	76.2%	30.9%
Female	57.2%	54.3%
Race/ethnicity – White	81.9%	67.4%
Race/ethnicity – Hispanic	4.4%	13.7%
Race/ethnicity – Black	8.9%	10.8%
Race/ethnicity – Asian	2.6%	5.7%
Race/ethnicity – AIAN / other / multiple	2.3%	2.5%
Ever smoked (former or current)	82.2%	37.5%
Current smoker	21.1%	12.0%
Ever used an e-cigarette	17.4%	16.1%
COPD	46.6%	5.5%
Heart disease	22.6%	6.0%
Hypertension	65.3%	36.5%
Diabetes	21.2%	10.7%
Asthma	18.8%	13.9%
Comorbidity count, mean	1.7	0.7
Fair or poor self-rated health	57.4%	15.1%
Underweight	5.4%	1.6%
Obese	27.2%	32.4%
Less than high school	14.1%	8.6%
Bachelor or graduate degree	18.3%	38.0%
Family income below the poverty line	15.3%	10.1%
Region – Northeast	18.9%	16.6%
Region – Midwest	28.1%	22.0%
Region – South	37.0%	36.2%
Region – West	16.0%	25.1%

2.6 Initial data-quality issues and limitations

- **Severe imbalance:** 619 cases out of 149,928 adults (0.41%).
- **Hidden missing values:** NHIS stores "refused / not ascertained / don't know" as numbers (7, 8, 9 or 97, 98, 99) that look like real values; 4,424 such codes sat in just four key columns.
- **A renamed column:** education is `EDUC_A` in 2019–2020 but `EDUCP_A` from 2021 (same codes), and code 11 (doctorate) exists only in 2019–2020.
- **Self-reported, lifetime answers:** someone diagnosed years ago may have quit smoking or lost weight since, so current habits can be a result of the cancer (reverse causation).
- **No smoking dose:** NHIS has no pack-years or quit dates, so we can only approximate the official screening group.
- **Survey design:** NHIS is not a simple random sample. It over-samples some groups – the sample is older than the population (31.1% aged 65+ vs 21.9% after weighting) – so raw rates (0.41%) differ from population rates (0.31%).
- **Year comparability:** the 2019 questionnaire redesign and the 2020 pandemic disruption could affect comparability between years; the year-to-year differences in the case rate are not significant ($p = 0.096$).
- **Coarse regional data:** each regional variable has only 20 distinct values (4 regions $\times$ 5 years), so it is ecological context, not personal exposure.
- **Possible repeats across years:** people cannot be linked between survey years, so a few adults may appear in two different years.

2.7 The dataset we rejected

We started the project with a 22,457-row file (`lung_cancer_combined_22457.csv`, no documented origin) that has the same columns as the well-known 309-person Kaggle lung-cancer survey [5]. Before using it we audited it the way we would review an unknown dependency, and it failed review (Table 14). The decisive evidence is the row order and the lineage; duplicates and label conflicts alone would be weak evidence, because a file with about fifteen low-cardinality columns produces repeated answer patterns naturally.

Table 14. Integrity audit of the supplied file (all numbers recomputed from the file), strongest evidence first.

Check	Finding	Why it matters
Row order	the first 1,455 rows are all negative; the first positive label appears at row 1,456	row position (and `patient_id`) predicts the label – a leakage path
Lineage	only 1,438 rows (6.4%) match a row of the public 309-person survey (252 of its patterns, one repeated 59 times; labels agree in 99.4% of matches); the other 21,019 rows match no public source	most of the file has no traceable origin
Signal after deduplication	no feature is associated with the label (all Cramér's $V < 0.02$)	the apparent signal came from the duplicates and the ordered block
Exact duplicates	2,573 rows repeat another row exactly	the same answers counted many times inflate every statistic
Label conflicts	133 identical answer patterns carry both labels (315 rows)	the label is not a clean function of the answers

This pattern is statistically consistent with resampling or programmatic generation; it is not proof that the data are synthetic. We replaced the file with NHIS and kept it in `data/raw/other/`

so the decision stays auditable. The lineage numbers were recomputed against an identical public copy of the Kaggle survey [5].

3. Preprocessing Methodology

We lost only 292 of 150,220 rows (0.19%), all of them people whose cancer answer was unknown, and ended with 0 missing values in the model-ready file.

3.1 A logged, reproducible pipeline

We built the cleaning as a pipeline in Python (pandas, SciPy [13], scikit-learn [14], statsmodels [15]) rather than as notebook cells run by hand:

- each step lives in a script – `build_regional_data.py` for the three regional sources and `build_clean_dataset.py` for NHIS;
- each step records its effect (rows, columns, missing cells) in `data/clean/before_after_summary.csv` and explains itself in `data/clean/cleaning_report.txt`;
- the whole project runs from one entry point, `python run_all.py` (Table 42), with every random step seeded (42).

That made every decision below (Table 15) easy to review and easy to undo, and it guarantees that the numbers in this report are the numbers the code produces.

3.2 Problems found and treatments applied

Table 15 lists every data-quality problem we found, what we did about it, why, and its measured effect.

3.3 Missing values

Before filling any gap (Fig. 2) we checked whether missingness itself carries information about the target. For every column with at least 1% missing, we cross-tabulated "value missing" against lung cancer with a χ^2 test (all expected counts ≥ 14.5, so the test is valid).

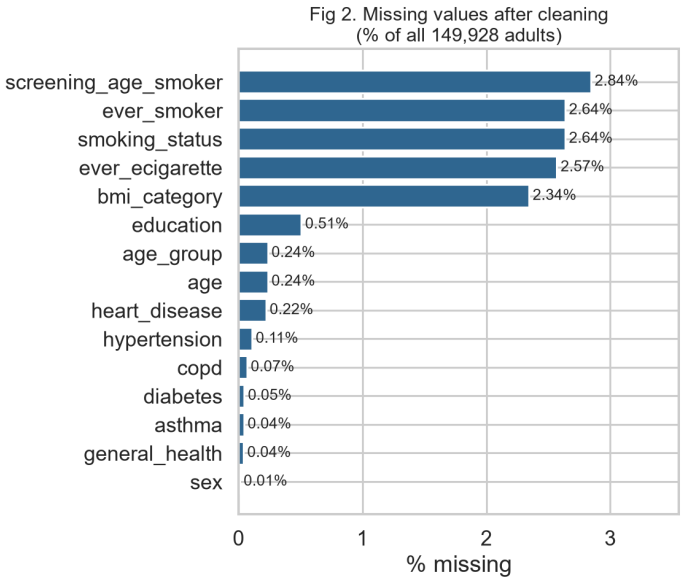


Figure 2. Share of missing values per column after recoding the hidden "unknown" codes. The five smoking, e-cigarette and BMI columns are the only ones above 2.3% (2.34–2.84%); every other column is below 0.6%.

Table 15. Data-quality problems, the treatment applied, the reason, and its measured effect.

Problem found	What we did	Why	Effect
Five yearly files; education renamed `EDUCP_A` from 2021; doctorate coded 11 only in 2019–2020	stacked the years, mapped both names to one column, mapped code 11 to "Graduate"	same codes every year, so one schema works	150,220 rows × 20 columns
Possible duplicate people	built `person_id` = year + household number and checked for repeats	NHIS interviews one adult per household per year, so the key is unique by design; the check guards against a stacking error rather than finding real duplicates	0 duplicates found
No ready-made target; `LUNGCAN_A` only asked to people who had any cancer	0 = no cancer or another cancer; 1 = lung cancer; refused / don't know → row removed	"don't know" is not "no", and we never impute a target	292 rows removed; 149,928 kept, 619 cases
Missing values hidden as numbers (7/8/9, 97/98/99)	recoded every column to readable labels; those codes → missing	otherwise a model would read 97 as a real value	4,424 hidden codes in 4 key columns became missing
Inconsistent categories (e.g. smoking codes 1, 2, 5 all mean "current")	merged into one label per meaning (Current / Former / Never)	one category per real-world meaning	3 clean smoking groups
Regional sources at a finer level (monitor, state) than NHIS (region)	aggregated to 4 regions × 5 years, then a left join on region + year	public NHIS files only identify the region	20 region-years; 149,928 rows before and after; 0 unmatched
Outliers	IQR rule on the 6 numeric columns plus a plausibility check of every flagged value (§3.4)	remove what is impossible, not what is rare	0 removed
Remaining missing values (at most 2.84%, Fig. 2)	categories → explicit Unknown; yes/no → 0 plus an `_unknown` flag; age → training median (54) plus a flag	keep every row; no detectable link between missingness and the target (Table 16)	0 missing cells in the model-ready file
Features that are exact or near-exact combinations of several other features (`comorbidity_count` = sum of 5 flags; complete one-hot groups summing to 1; region dummies vs the regional columns) – invisible to a pairwise correlation filter	Rule 4 of feature selection: variance inflation factor (VIF) on training rows; dropped `comorbidity_count` (kept for the EDA), one reference category per one-hot group and the region dummies	a linear model cannot be fitted with perfect multicollinearity, and its coefficients are unstable with near-perfect multicollinearity	VIF: 28 of 48 features infinite → maximum 7.5 (none above 10); 48 → 38 features
Leakage risk	never used `CANEV_A` ("ever had any cancer") as a feature; split before fitting anything	every case has `CANEV_A` = yes, so it would reveal the answer	no feature derived from the target

Table 16. Is missingness related to lung cancer? χ^2 tests of "value missing (yes/no)" × target.

Column	% missing	Rate if missing	Rate if present	χ^2	p	Holm p
bmi_category	2.34	0.37%	0.41%	0.07	0.79	1.00
ever_ecigarette	2.57	0.44%	0.41%	0.02	0.88	1.00
smoking_status / ever_smoker	2.64	0.46%	0.41%	0.09	0.77	1.00
screening_age_smoker	2.84	0.45%	0.41%	0.05	0.83	1.00

None of the differences is significant (all $p > 0.7$; Table 16), so there is no detectable link between the gaps and lung cancer. We word this carefully: each missing group contains only about 14–18 expected cases, so the tests have little power, and they do not check whether missingness depends on other variables such as age or education. With at most 2.84% missing per column, filling the gaps cannot move the main estimates much. We still keep an Unknown category or an `_unknown` flag wherever the gap is large enough to be informative, so a future model can learn from "missing" if it matters. Imputation values (the age median) are computed on training rows only (Section 3.7).

3.4 Outliers

The IQR rule flags values that are unusual, not values that are wrong (Fig. 3, Table 17). Removing the 7,451 adults with three or more chronic conditions would delete exactly the sick, older people among whom lung cancer is concentrated, and would bias every later estimate. Our rule was therefore to remove only impossible values; there were none.

Fig 3. Outlier check (IQR rule): every flagged value is plausible, so all values are kept

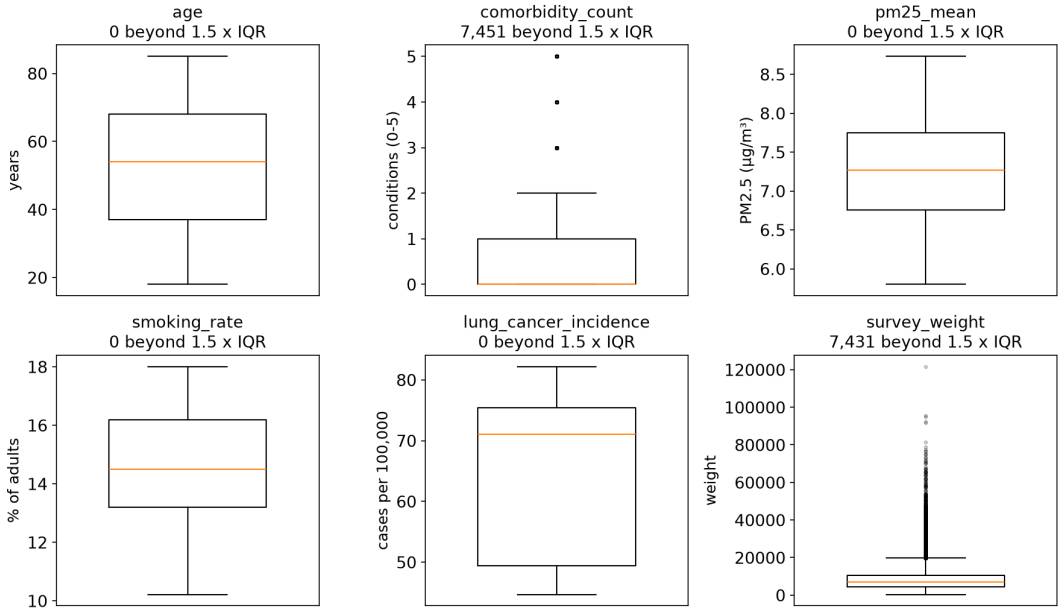


Figure 3. Box plots of the six numeric columns with the number of values beyond 1.5× IQR. Every flagged value was checked for plausibility and kept.

3.5 Integrating the regional sources

Each regional source needed its own cleaning before the three could be combined (`build_regional_data.py`; Table 18):

We mapped every state to its US Census region – the same four regions NHIS uses – and combined the three tables into one 20-row table (`data/clean/regional_context.csv`). No region-year was missing; the code would fill a missing year from the same region's nearest year and log it. The NHIS table was then left-joined on region + year with a many-to-one validation, and an assertion stops the pipeline if the join ever adds a row. Result: 149,928 rows before and after, 0 adults without a regional match.

Two aggregation choices are worth stating plainly because they are limitations: the registry rate is a crude rate (not age-adjusted), and states are averaged without population weights. Both are discussed in Section 4.8.

Table 17. Outlier check (IQR rule: below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$) and the decision for each column.

Column	Bounds	Flagged	Decision and reason
age	[−9.5, 114.5]	0 (0%)	keep – every age 18-85 is a real adult; 85 means "85 or older" (NHIS top-code), not an error
comorbidity_count	[−1.5, 2.5]	7,451 (4.97%)	keep – a count from 0 to 5; high values are real people with several diseases
pm25_mean	[5.3, 9.2]	0 (0%)	keep – official EPA measurements; only 20 region-year values, none impossible
smoking_rate	[8.7, 20.7]	0 (0%)	keep – official CDC percentages between 0 and 100
lung_cancer_incidence	[10.4, 114.4]	0 (0%)	keep – official registry rates; differences between regions are real
survey_weight	[−4883.8, 19849.6]	7,431 (4.96%)	keep – NHIS design weight, used for population estimates only, not as a feature

Table 18. Cleaning and aggregation of the three regional sources.

Source	Cleaning steps	Result
EPA PM2.5	kept the parameter "PM2.5 – Local Conditions" and the official annual standard; kept "No Events" / "Events Excluded" rows (wildfire-type events removed); dropped values outside 0–100 µg/m^3 and duplicate monitors; dropped 56 rows outside the 50 states + DC; averaged per state first (so states with many monitors do not dominate), then per region	1,409–1,798 annual monitor values per year → 20 region-year means
BRFSS smoking	kept the "current smoker = yes" answer, years 2019–2023, values 0–100%; dropped 21 territory / national rows; averaged per region	1,644 rows → 251 state-years → 20 region-years
US Cancer Statistics	kept lung cancer, both sexes, all races, single years (multi-year rows dropped); suppressed counts treated as missing; region rate = sum of cases ÷ sum of population × 100,000	23,796 rows → 255 state-years → 20 region-years

3.6 Effect of each step

Table 19 follows the data through the pipeline. The column count changes for simple reasons: step 2 adds `person_id` (20 → 21), step 3 adds the target (22), and step 4 replaces the raw NHIS columns with 19 readable ones (the helper columns `HHX`, `CANEV_A` and `LUNGCAN_A` are not carried forward). The missing cells fall sharply at step 4 because `LUNGCAN_A`, blank by design for everyone without cancer, is no longer carried; they rise again at step 5 only because the new columns inherit the gaps of the columns they are built from. The 42 columns of the model-ready file are the 38 features plus `person_id`, `split`, `survey_weight` and the target.

Table 19. Rows, columns and missing cells after each preprocessing step (`data/clean/before_after_summary.csv`).

Step	Rows	Columns	Missing cells	What changed
1. Load + stack 5 years	150,220	20	131,400	5 yearly files stacked; 20 needed columns kept (LUNGCAN_A is blank by design for people who never had cancer)
2. Remove duplicates	150,220	21	131,400	0 duplicate person IDs removed
3. Build target	149,928	22	131,217	292 rows with unknown answer removed
4. Recode codes → values	149,928	19	13,235	number codes turned into words; e.g. 4,424 "unknown" codes in 4 key columns → missing
5. New columns	149,928	23	21,805	added age_group, ever_smoker, screening_age_smoker, comorbidity_count
5b. Join regional table	149,928	26	21,805	3 columns added (pm25_mean, smoking_rate, lung_cancer_incidence); 0 rows added
5c. Outlier check	149,928	26	21,805	IQR check on 6 numeric columns; all values plausible → kept
6. Checks + save clean file	149,928	26	21,805	0 duplicate IDs, ages 18-85, missing values listed
7. Encode + fill gaps (train only)	149,928	71	0	one-hot encoding, training-median filling, "unknown" flags → 0 missing
8. Feature selection	149,928	42	0	29 redundant / near-constant / near-duplicate / linearly dependent features dropped; 38 kept (max VIF 7.5)
9. Scaling (train only)	149,928	42	0	4 numeric columns standardised

3.7 How we prevented data leakage

We treated the test set like production data the model has never seen, following the leakage checklist of Kapoor and Narayanan [11]:

- The 80/20 split (stratified on the target, seed 42) happens before anything is learned from the data: 119,942 training rows and 29,986 test rows, both with a 0.41% case rate.
- The age median used for filling, the feature-selection rules and the VIF step, the scaler and the PCA are all computed on training rows only, then applied unchanged to the test rows.
- `CANEV_A` is never a candidate feature, `person_id` is an identifier only, and `survey_weight` is used for population estimates, never as a feature.
- A sanity check: after scaling, the training means are exactly 0 while the test means are slightly off (−0.002 for age, 0.006 for PM2.5, −0.001 for the regional smoking rate and 0.019 for registry incidence; Table 33). That is exactly what we expect if the scaler never saw the test rows, and `supplementary_analysis.py` turns it into an assertion: the saved scaler parameters must equal a refit on training rows only and must differ from a fit on all rows.

Descriptive statistics, the population profile and the hypothesis tests in Sections 2.5 and 4 use all adults, because they describe the population and fit nothing that a model reuses. Two design decisions were nevertheless informed by data on all rows, and we state them openly: merging the

three smallest race groups for the χ^2 test (a testing decision only; the model-ready file keeps all seven groups, with White as the reference), and the `screening_age_smoker` feature, which comes from the USPSTF rule [6] and was only illustrated by Fig. 8. The one EDA step that ranks features (mutual information, §5.5) uses training rows only.

4. Exploratory Data Analysis and Results

Age, smoking history and COPD separate people with and without lung cancer far more than any regional factor does. Fourteen of our eighteen tests stay significant after correction, but the effect sizes remind us that with 150,000 rows "significant" and "important" are not the same thing. For every test we followed the same checklist: state H_0, check the assumptions, run the test, report an effect size, and say what it means in plain words. Unless stated otherwise, p-values in the text are raw (before correction); Holm-adjusted values are always labelled "Holm p".

4.1 Descriptive statistics and univariate analysis

The big picture. 149,928 adults; mean age 52.8 years (SD 18.4), median 54 (IQR 37–68); 54.3% women; 619 lung-cancer cases (0.41% of the sample, 0.31% once we apply the NHIS survey weights). The gap between the two rates is not a bug: the sample is older than the US population, so we use weighted rates whenever we talk about the population. The composition of the sample by every variable is described in Section 2.5.

- **Age is not normal.** It is flat with two humps (negative excess kurtosis, −1.08; Table 20) and a spike at the top-code. Both normality tests reject normality: Shapiro–Wilk on a random sample of 5,000 ages, $p = 5.7 \times 10^{-33}$, and D'Agostino's K^2 on all adults, $p < 10^{-300}$. The Q-Q plot confirms it visually (Fig. 4). We therefore use rank-based methods (Mann–Whitney, Spearman) for age, as the course recommends when normality fails (Chapter 4).
- **Comorbidity count is right-skewed** (skewness 1.29): half of adults have none of the five conditions, 31.7% have one, and only 5.0% have three or more.
- **No transformation is needed.** The only skewed column is a small count from 0 to 5, not a long-tailed amount such as income, so a log transform would not help; the regional columns are roughly symmetric.

Table 20. Centre, spread and shape of the numeric columns (all 149,928 adults with the value known, unweighted).

Column	Mean	Median	SD	IQR	Min	Max	Skewness	Excess kurtosis
age (years)	52.8	54.0	18.4	31.0	18.0	85.0	−0.05	−1.08
comorbidity_count	0.73	0.00	0.91	1.00	0.00	5.00	1.29	1.54
pm25_mean (µg/m^3)	7.29	7.27	0.76	0.99	5.80	8.73	−0.06	−0.71
smoking_rate (%)	14.47	14.49	2.08	3.00	10.22	18.01	−0.00	−0.83
lung_cancer_incidence (/100k)	67.4	71.0	12.5	26.0	44.7	82.2	−0.87	−0.81

4.2 RQ1 – Smoking

- **Hypotheses.** H_0: smoking status and lung cancer are independent; H_1: they are associated.
- **Test and assumption.** Both variables are categorical (a 3×2 table), so we use Pearson's χ^2 test of independence. Its assumption holds: the smallest expected count is 72.4, far above 5.
- **Result.** $\chi^2(2) = 528.0$, $p = 2.2 \times 10^{-115}$ (Holm $p = 3.1 \times 10^{-114}$), Cramér's $V = 0.060$ – small by Cohen's conventions [9].

Fig 4. Distribution of the numeric variables
(adults with the value known: age n = 149,575, comorbidity count n = 149,928)

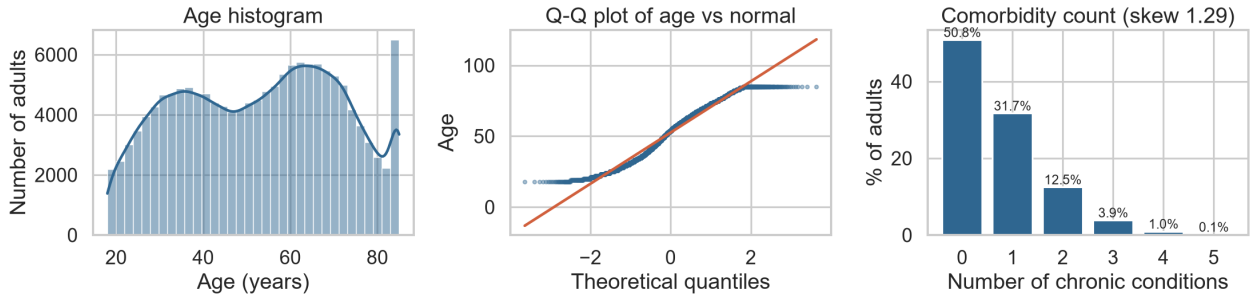


Figure 4. Numeric variables. Left: age histogram (two humps, and a pile-up at 85 = "85 or older"). Centre: Q-Q plot of age against a normal distribution (the S-shape of a flat distribution). Right: comorbidity count (right-skewed).

Fig 5. Who is in the data? Categorical variables
(unweighted; % of the adults with a known answer, out of 149,928)

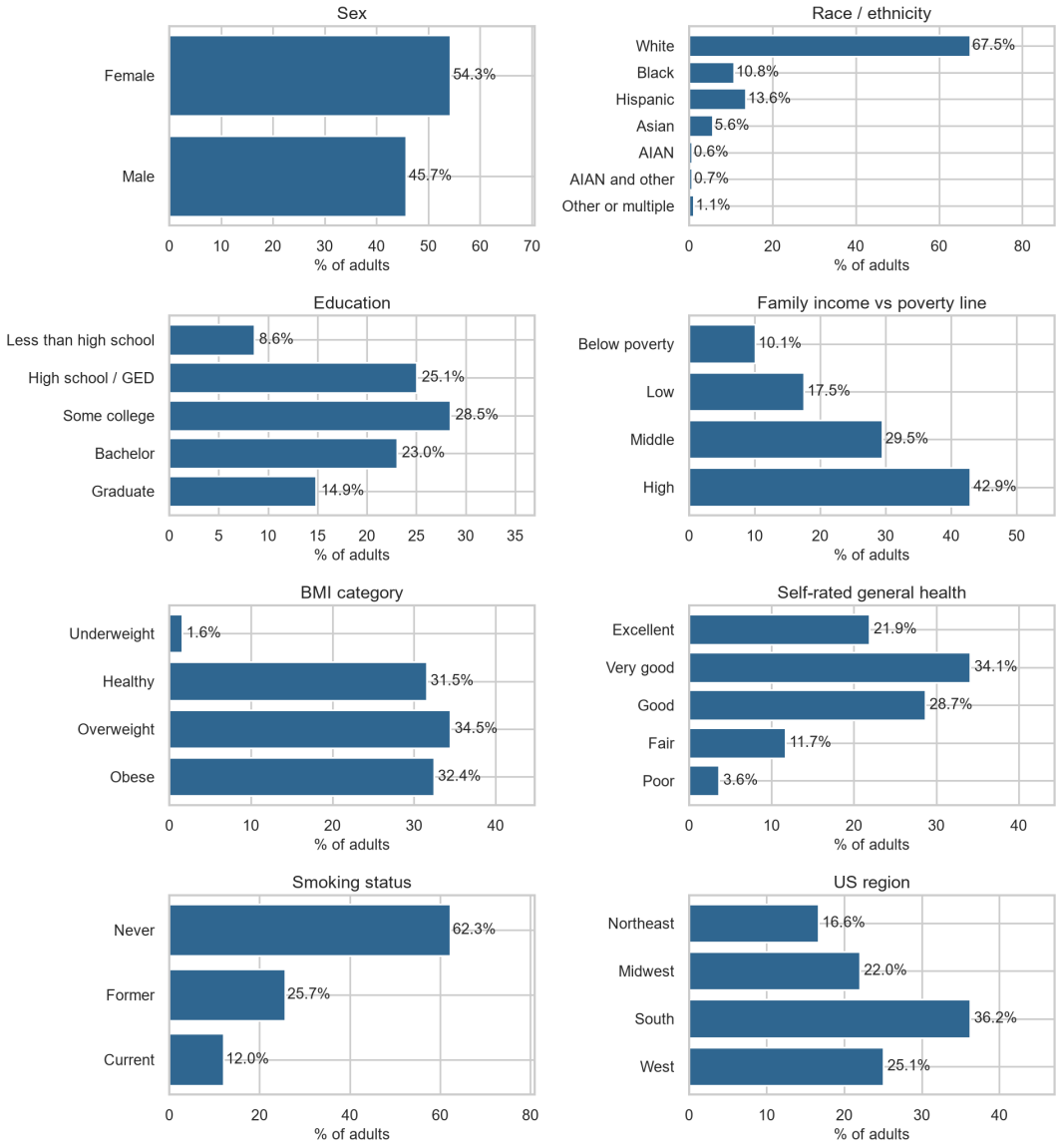


Figure 5. Distribution of the categorical variables (unweighted shares of adults with a known answer). Counts, cases and rates for every category are in Tables 8–12.

Table 21. Lung-cancer rate by smoking status (absolute risk) and risk ratio against never-smokers (adults with known smoking status).

Smoking status	Adults	Cases	Sample rate	Weighted (population) rate	Risk ratio (weighted)
Never	90,937	107	0.12%	0.08%	1 (reference)
Former	37,452	367	0.98%	0.81%	10.0
Current	17,586	127	0.72%	0.60%	7.4

- **Effect size.** Ever-smokers vs never-smokers: odds ratio 7.66 (95% CI 6.22–9.44). In absolute terms, about 8 in 10,000 never-smokers have a lung-cancer history in the population, against about 81 in 10,000 former and 60 in 10,000 current smokers (Table 21, Fig. 6).
- **How we read it.** The tiny p-value mostly reflects our sample size, and V looks small because 99.6% of every group is cancer-free – V is bounded by the rarity of the outcome. The odds ratio is the meaningful effect size for a rare outcome: ever-smokers have about 7.7 times the odds of a lung-cancer history.
- **Why former smokers look worse than current ones (0.81% vs 0.60%).** At first this looked like a bug, but it is not one. NHIS records smoking now, and many people quit after their diagnosis, which moves them into the "former" group (reverse causation). Former smokers are also older (median age 63 vs 52 for current smokers), which is why we look at age next.

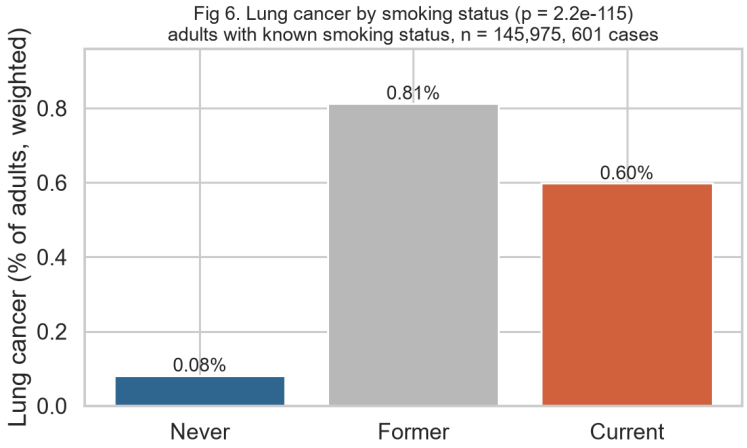


Figure 6. Weighted share of adults with a lung-cancer history, by smoking status (the p-value in the title is the raw χ^2 p-value).

4.3 RQ2a – Age, and how it interacts with smoking

- **Difference.** The median age is 72 with lung cancer vs 54 without, and the weighted rate rises steadily from the youngest to the oldest age group (Fig. 7).
- **Test choice.** H_0: both groups have the same age distribution. Age is not normal (§4.1) and the groups are very unequal (618 vs 148,957 adults with a known age), so we use the Mann–Whitney U test rather than a t-test.
- **Result.** $U = 73{,}126{,}291$, $p = 3.1 \times 10^{-141}$, rank-biserial $r = 0.59$ – a large effect. In plain words: if we pick one adult with and one without a lung-cancer history at random, the one with lung cancer is older $(1 + 0.59)/2 \approx 79.5\%$ of the time.
- **Robustness.** With samples this large the t-test is also valid (central limit theorem), so we ran Welch's t-test as a check: $t = 44.9$, $p = 4.6 \times 10^{-199}$. Both tests agree, so the conclusion does not depend on the choice of test.

Fig 7. Lung cancer increases with age (Mann-Whitney p = 3.1e-141; n = 149,575 adults with known age, 618 cases)

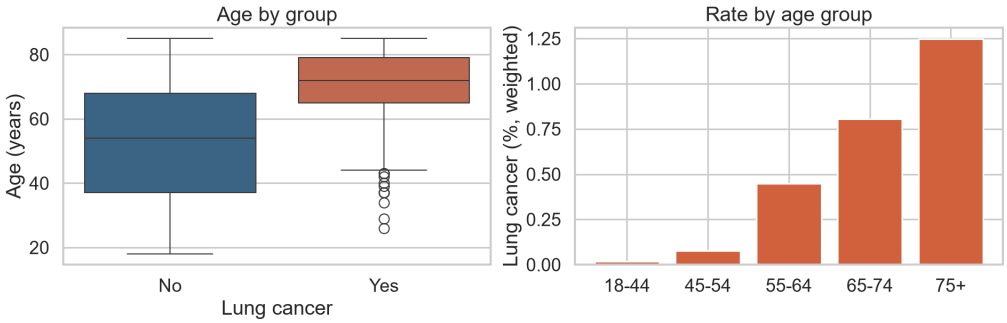


Figure 7. Left: age distribution with and without lung cancer. Right: weighted lung-cancer rate by age group.

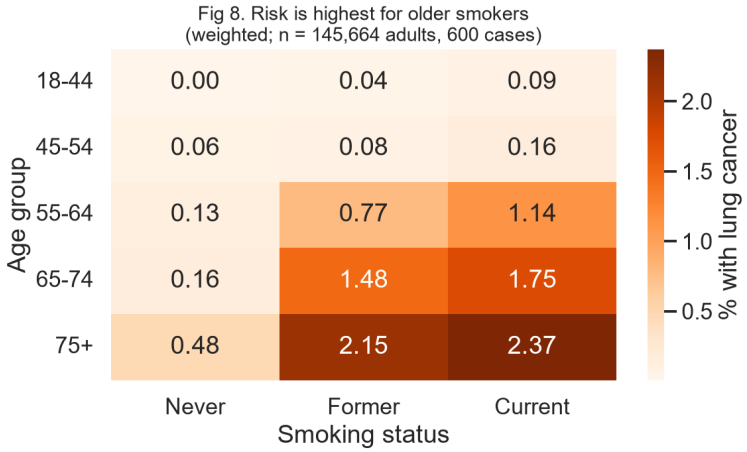


Figure 8. Weighted lung-cancer rate (%) for each age group and smoking status. Cells for adults under 45 contain very few cases (3 never-smokers and 10 ever-smokers in total), so their rates are imprecise.

- **Age and smoking act together.** Never-smokers stay below 0.5% even at 75+. Ever-smokers go from under 0.1% before 45 to 2.15% (former) and 2.37% (current) at 75+.
- **Smoking is not just age in disguise – tested.** Because ever-smokers are older, part of the crude odds ratio could be age. We therefore compared ever- with never-smokers inside each age group and pooled the groups with the Cochran–Mantel–Haenszel (CMH) method (Table 22). The odds ratio falls from 7.6 to 5.8 (95% CI 4.7–7.1) once age group is held fixed, and the CMH test rejects "no association within age groups" ($\chi^2 = 340.5$, $p = 5 \times 10^{-76}$). Part of the crude effect was therefore age, but a strong smoking association remains. The Breslow–Day test ($p = 0.005$) shows the size of the effect differs between age groups: it is about 8 from 55 to 74, 4.4 at 75+, and not significant at 45–54 (only 21 cases), so the pooled 5.8 is an average.
- This combination of age and smoking is exactly the logic of the USPSTF screening rule, which combines age 50–80 with smoking history – and the reason we built the `screening_age_smoker` feature (§5.1, Table 32).

Table 22. Smoking and lung cancer inside each age group (ever vs never smokers; adults with known smoking status and age). CMH $\chi^2 = 340.5$, $p = 5 \times 10^{-76}$; Breslow–Day $p = 0.005$. Source: `smoking_by_age_group.csv`.

Age group	Adults	Cases, ever-smokers	Cases, never-smokers	Odds ratio [95% CI]
18–44	53,022	10	3	7.7 [2.3, 25.9]
45–54	20,987	11	10	1.9 [0.8, 4.3]
55–64	26,358	96	15	8.2 [4.8, 14.0]
65–74	25,842	187	27	7.9 [5.3, 11.7]
75+	19,455	189	52	4.4 [3.2, 5.9]
Crude (age ignored)	145,664			7.6 [6.2, 9.4]
Mantel–Haenszel (adjusted for age group)	145,664			5.8 [4.7, 7.1]

4.4 RQ2b – Chronic conditions

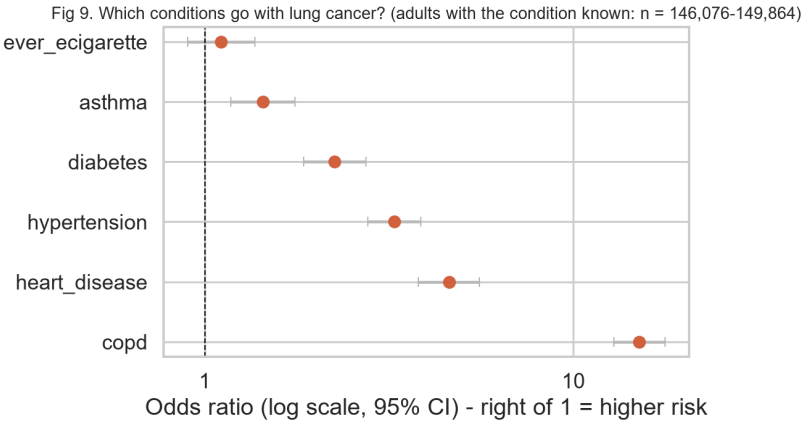


Figure 9. Odds ratio of a lung-cancer history for each condition, with 95% confidence intervals (log scale). Right of 1 means higher odds.

- **COPD is the strongest signal in the whole project.** It shares smoking as a cause and is also a known risk factor on its own – which is why we test it against smoking (and age) in §4.5.
- **Heart disease, hypertension and diabetes** are partly proxies for age; these odds ratios (Fig. 9) are not age-adjusted, so we do not over-read them.

Table 23. Chronic conditions and e-cigarette use vs lung cancer (χ^2 tests; all expected counts ≥ 5).

Condition	Rate if yes	Rate if no	Odds ratio [95% CI]	Cramér's V	Holm p
COPD	3.41%	0.23%	15.1 [12.9, 17.7]	0.114	$< 10^{-300}$
Heart disease	1.53%	0.34%	4.6 [3.8, 5.6]	0.045	7.3×10^{-65}
Hypertension	0.74%	0.23%	3.3 [2.8, 3.9]	0.038	1.5×10^{-48}
Diabetes	0.82%	0.36%	2.3 [1.9, 2.7]	0.022	6.7×10^{-16}
Asthma	0.56%	0.39%	1.4 [1.2, 1.8]	0.009	2.9×10^{-3}
E-cigarette (ever)	0.45%	0.41%	1.1 [0.9, 1.4]	0.002	1.00

- **Asthma** survives correction, but its effect is negligible ($V = 0.009$): a clear example of significant without being important.
- **E-cigarette use shows nothing** (raw $p = 0.39$, Holm $p = 1.00$). That makes sense: e-cigarettes are recent, users are mostly young (mean age 41 vs 55 for non-users, Table 11), and lung cancer takes decades to develop. Section 4.10 reports what this null result can and cannot rule out.

4.5 RQ2b – Is the COPD link only due to smoking?

Smokers develop COPD more often and lung cancer more often, so smoking alone could create a COPD–lung-cancer association – a classic confounder. To separate the two, we compared adults with and without COPD within each smoking group (Fig. 10, Table 24), then pooled the three groups with the Cochran–Mantel–Haenszel (CMH) test [16], which tests H_0: "no COPD–lung-cancer association once smoking status is held fixed".

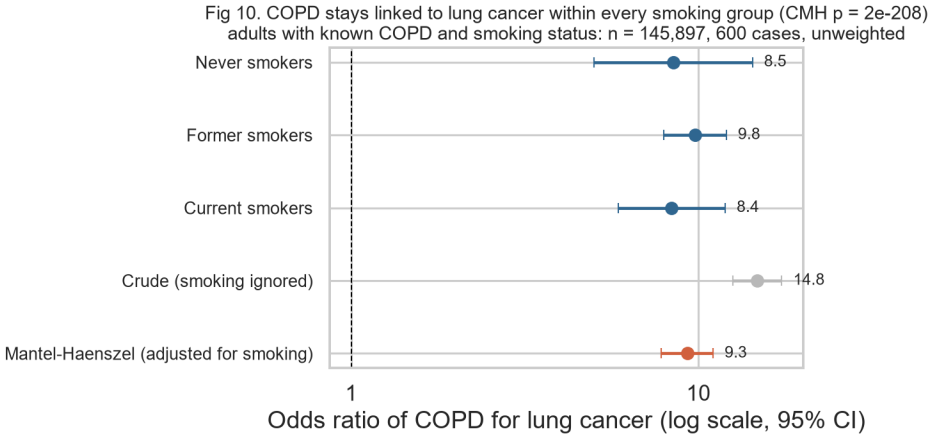


Figure 10. Odds ratio of COPD for lung cancer within each smoking group, ignoring smoking (crude) and adjusted for smoking (Mantel–Haenszel), with 95% confidence intervals (log scale).

Table 24. COPD and lung cancer, stratified by smoking status (adults with known COPD and smoking status).

Group	Adults	Cases	Odds ratio [95% CI]
Never smokers	90,901	107	8.5 [5.0, 14.3]
Former smokers	37,424	366	9.8 [7.9, 12.0]
Current smokers	17,572	127	8.4 [5.9, 11.9]
Crude (smoking ignored)	145,897	600	14.8 [12.6, 17.4]
Mantel–Haenszel (adjusted for smoking)	145,897	600	9.3 [7.8, 11.0]

- **CMH test:** $\chi^2 = 949.7$, $p = 1.5 \times 10^{-208}$ (Holm $p = 2.4 \times 10^{-207}$) – the association survives adjustment for smoking.
- **Breslow–Day test** of equal odds ratios across the three groups: $p = 0.69$, so the effect is consistent across groups and pooling them is justified.
- **How we read it.** The crude odds ratio of these 145,897 adults is 14.8 (slightly below the 15.1 of Table 23, which uses everyone with a known COPD answer); it falls to 9.3 once smoking is held fixed, so part of the crude association was due to smoking. But COPD remains strongly associated with lung cancer in never, former and current smokers alike – even never-smokers with COPD have 8.5 times the odds.
- **Age as a second confounder.** Adults with COPD are much older (mean age 65.4 vs 52.0). On the 145,587 adults whose age is also known (599 cases), the crude odds ratio is 14.7 and the smoking-adjusted one 9.2 – the same picture as above – and stratifying by smoking status and age group together (15 strata; Table 25) lowers it further, to 5.7 (95% CI 4.8–6.8), still highly significant ($\chi^2 = 501.9$, $p = 3.7 \times 10^{-111}$). COPD is therefore strongly associated with lung cancer beyond what smoking and age explain – still an association, not a cause.

Table 25. COPD odds ratio under increasing adjustment (the 145,587 adults with known COPD, smoking status and age; 599 cases). Source: `copd_adjusted.csv`.

Adjusted for	Odds ratio [95% CI]
None (crude)	14.7 [12.5, 17.3]
Smoking status (3 strata)	9.2 [7.8, 11.0]
Smoking status × age group (15 strata)	5.7 [4.8, 6.8]

4.6 RQ2c – Socio-demographic and health factors

Table 26. Socio-demographic and health factors vs lung cancer (χ^2 tests of independence). The three smallest race groups (AIAN, AIAN and other, Other or multiple; 2.5% of adults) were merged for the test, because their expected case counts fell below 5 (smallest 4.0), which makes the χ^2 approximation unreliable. After merging, the smallest expected count is 15.5. The model-ready file keeps all seven groups as categories (White is the reference category).

Factor	Weighted rate: highest → lowest group	χ^2 (df)	Holm p	Cramér's V
Sex	Female 0.32% → Male 0.30%	2.0 (1)	0.62	0.004
Race / ethnicity	White 0.39% → Hispanic 0.10%	70.7 (4)	1.1×10^{-13}	0.022
Education	Less than high school 0.49% → Bachelor 0.15%	114.7 (4)	7.8×10^{-23}	0.028
Income vs poverty line	Below poverty 0.46% → High 0.21%	78.0 (3)	7.5×10^{-16}	0.023
BMI category	Underweight 1.14% → Obese 0.24%	69.2 (3)	5.2×10^{-14}	0.022
Self-rated health	Poor 2.47% → Excellent 0.03%	1,154.7 (4)	1.8×10^{-247}	0.088

- **Sex** is not associated with lung cancer here (raw $p = 0.16$, Holm $p = 0.62$; Table 26).
- **A social gradient.** Less education and lower income go with higher rates (Fig. 11). Smoking follows the same gradient – 46% of adults without a high-school diploma have ever smoked, against 28% of bachelor's graduates, and 44% below the poverty line against 33% in the high-income group – so part of this gradient is the smoking effect again.
- **Underweight and poor self-rated health have the highest rates.** This is most likely reverse causation: cancer and its treatment cause weight loss and poor health, not the other way round. These features describe people who are already sick, which matters for the modelling phase (§5.7).

Fig 11. Lung cancer rate by socio-demographic and health factors
(survey-weighted; adults with the factor known, n = 146,414-149,928)

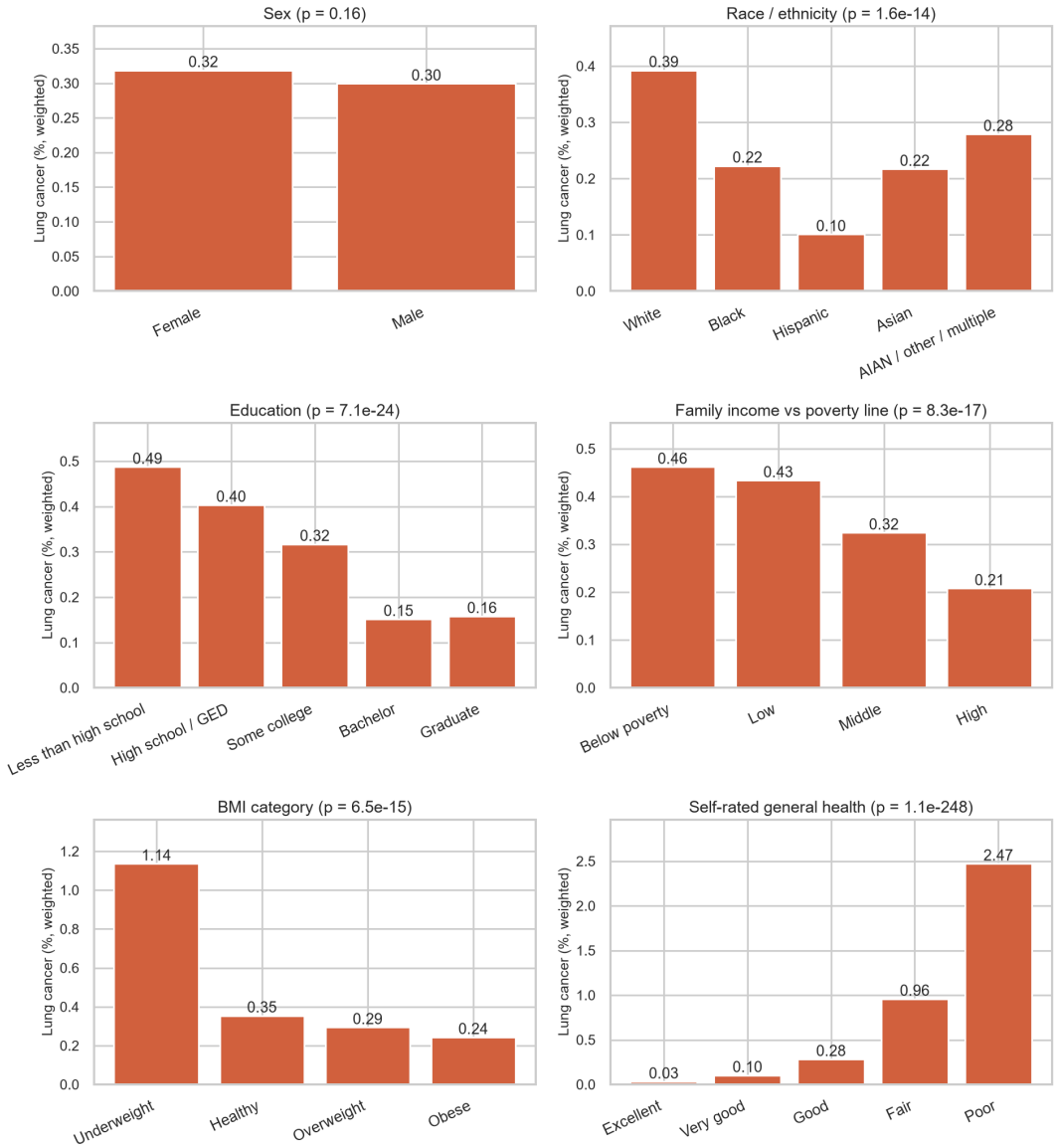


Figure 11. Weighted lung-cancer rate (%) by sex, race/ethnicity, education, income, BMI category and self-rated general health, with the raw χ^2 p-value of each test in the panel titles (Holm-adjusted values in Table 26).

- **Race differences** may partly reflect differences in age and smoking between groups (Table 9); they are not adjusted here. All Cramér's V values are small, again because the outcome is rare. These tests treat education, income, BMI and health as nominal; a trend test that uses their order would be more powerful, which can only strengthen these already significant results.

4.7 Correlations between variables

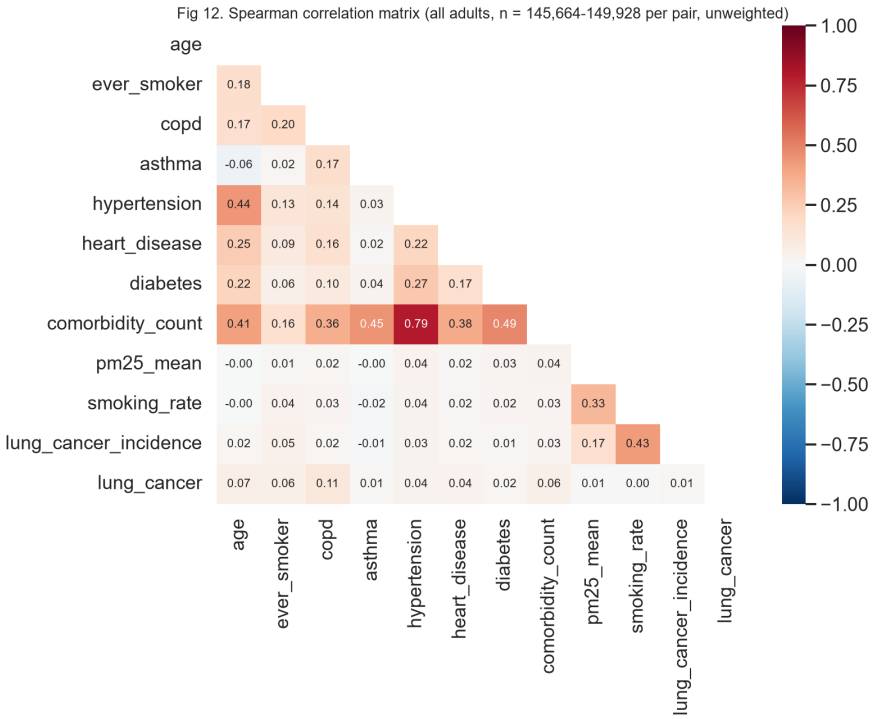


Figure 12. Spearman correlation matrix of the numeric and binary variables (lower triangle; all adults, unweighted; each pair uses the adults with both values known, n = 145,664–149,928).

- **Why Spearman.** Most columns are 0/1 or ordinal, and age is not normal; Spearman uses ranks and only assumes a monotonic relationship, whereas Pearson assumes linearity (Chapter 4).
- **The strongest pairs are expected.** Hypertension–comorbidity count (0.79), diabetes–comorbidity count (0.49) and asthma–comorbidity count (0.45) are high because the count is built from those columns. Age also goes with hypertension (0.44) and the comorbidity count (0.41), and the regional smoking rate with registry incidence (0.43).

Fig. 12 only holds the numeric and 0/1 columns, so it cannot show how the categorical variables relate to each other or to the target. Fig. 12b therefore covers *all* 24 analysis variables, with the measure that suits each pair of types, and Fig. 12c covers *all* 38 features handed to the modelling phase. Every pair above 0.5 in either matrix is listed in Table 27 with the reason it is high (`reports/tables/high_correlation_pairs.csv`).

- **Redundancy – what is high, and what we did about it.** All twelve pairs above 0.5 are either *built-from* (one column is computed from the other, or both are joined on region + year; 11 pairs) or *same-concept* (two flags for the same missing block; 1 pair). **No two separately measured personal variables are associated above 0.5.** The built-from overlaps are handled in feature selection: `ever_smoker` and the age bands are dropped by rule 1, and `comorbidity_count` and the region dummies by the new rule 4 (§5.4). The two high pairs left among the final features are kept on purpose: the two "unknown" flags, which together identify adults with an unknown smoking status (§5.7), and `screening_age_smoker`

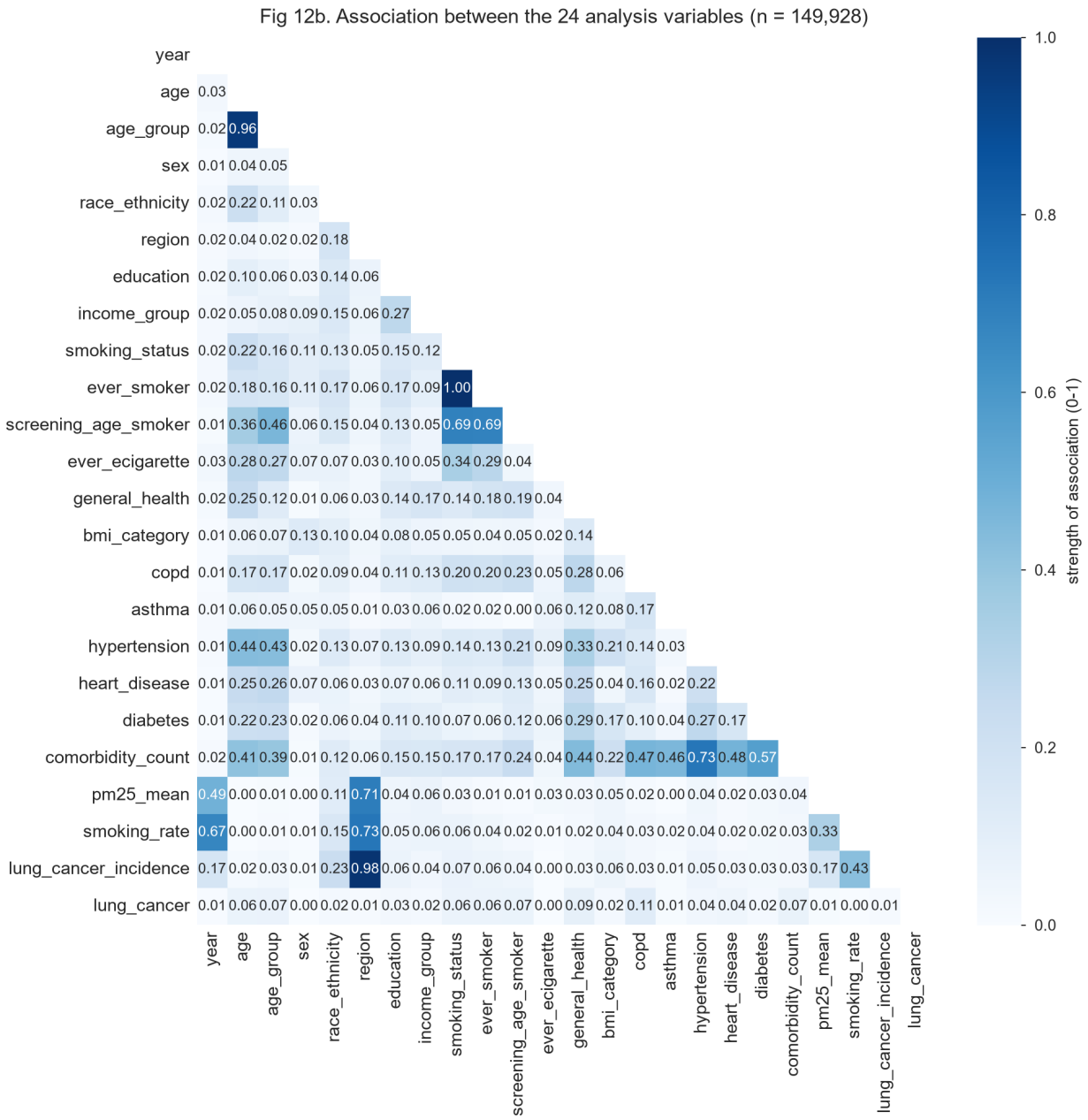


Figure 12b. Association between all 24 analysis variables of the clean dataset and the target (all 149,928 adults, 619 cases; each pair uses the adults with both values known, n = 142,511–149,928; unweighted). Cramér's V for two categorical variables, the correlation ratio η for a categorical and a numeric variable, |Spearman| for two numeric variables; all three run from 0 to 1. `person_id` and `survey_weight` are excluded, and `CANEV_A` is not in the clean dataset.

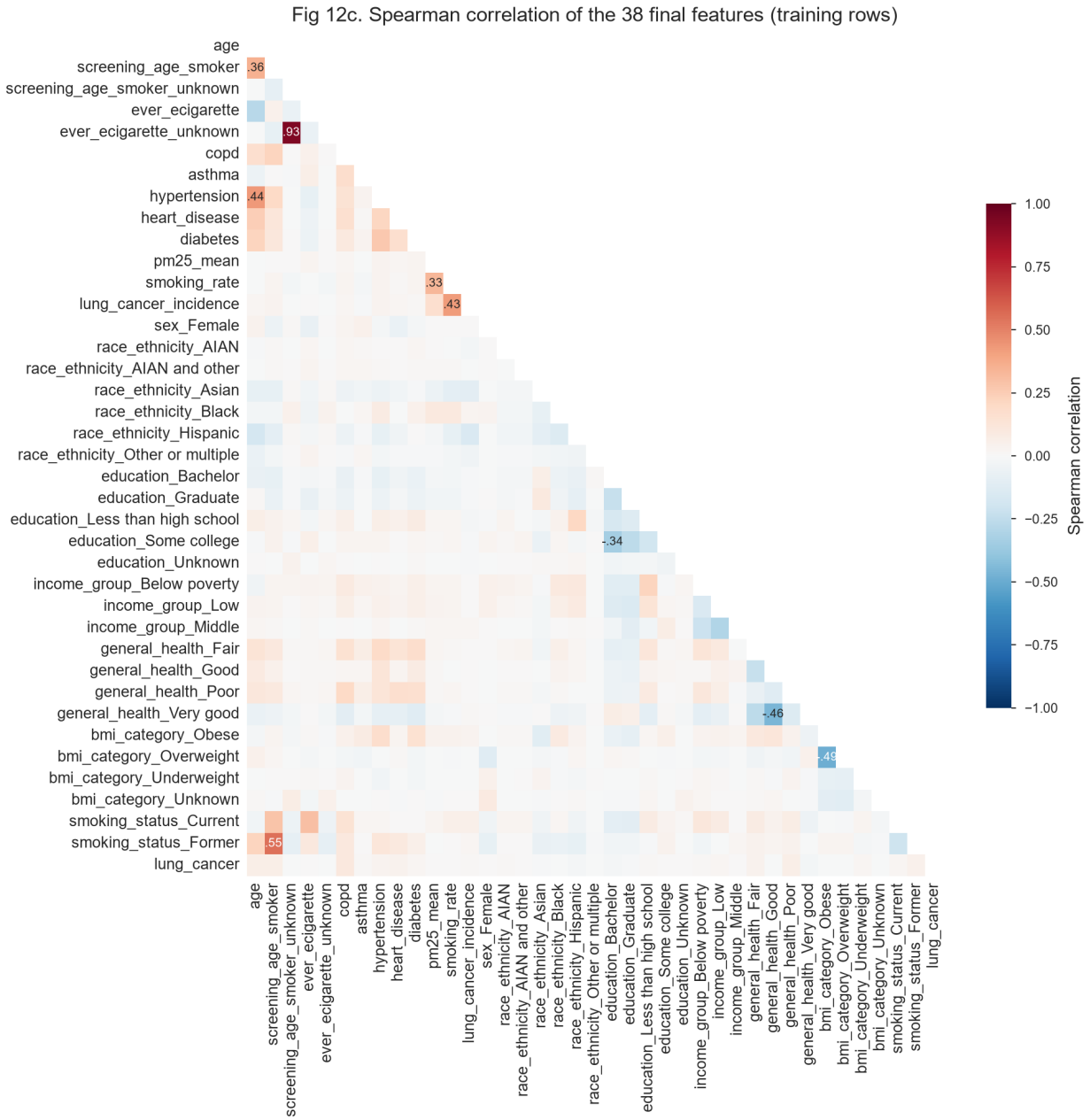


Figure 12c. Spearman correlation of the 38 final model features and the target, on the training rows only (n = 119,942 adults, 495 cases; unweighted; lower triangle; values printed where $|r| \geq 0.3$).

Table 27. Every pair of variables (Fig. 12b) or features (Fig. 12c) with an association above 0.5.

Matrix	Pair	Measure	Value	Type	Why it is high
Fig. 12b	year – smoking_rate	η	0.67	built-from	smoking_rate is joined on region + year, and BRFSS smoking fell every year 2019–2023
Fig. 12b	age – age_group	η	0.96	built-from	age_group is age cut into bands
Fig. 12b	region – pm25_mean	η	0.71	built-from	pm25_mean is joined on region + year (20 values)
Fig. 12b	region – smoking_rate	η	0.73	built-from	smoking_rate is joined on region + year (20 values)
Fig. 12b	region – lung_cancer_ incidence	η	0.98	built-from	lung_cancer_incidence is joined on region + year; the West is far lower than the other regions
Fig. 12b	smoking_status – ever_ smoker	Cramér's V	1.00	built-from	ever_smoker = Current or Former smoking_status
Fig. 12b	smoking_status – screening_ age_smoker	Cramér's V	0.69	built-from	screening_age_smoker = aged 50–80 AND ever smoked
Fig. 12b	ever_smoker – screening_ age_smoker	Cramér's V	0.69	built-from	screening_age_smoker = aged 50–80 AND ever smoked
Fig. 12b	hypertension – comorbidity_ count	η	0.73	built-from	comorbidity_count is the sum of the 5 condition flags, hypertension the most common (36.5%)
Fig. 12b	diabetes – comorbidity_count	η	0.57	built-from	comorbidity_count is the sum of the 5 condition flags
Fig. 12c	screening_age_smoker_ unknown – ever_ecigarette_ unknown	Spearman	0.93	same-concept	both flag the same ∼2.8% of adults who skipped the smoking / tobacco questions
Fig. 12c	screening_age_smoker – smoking_status_Former	Spearman	0.55	built-from	screening_age_smoker is built from smoking status, and most former smokers are aged 50–80

with former smoking, because the screening rule needs smoking status and age together. Both have a variance inflation factor below 10 (Table 35). Pairwise matrices also have a blind spot: a column can be an exact combination of *several* others (the comorbidity count is the sum of five flags; every complete one-hot group sums to 1) without any single pair exceeding 0.5. Hypertension–comorbidity count was the only visible trace of such a dependency in Fig. 12 (Spearman 0.79). This is why feature selection now also checks the VIF (§5.4).

- **Why every correlation with the target is small.** Nothing correlates strongly with the target (Fig. 12: highest COPD 0.11, age 0.07; Fig. 12b: COPD 0.11, general health 0.09). This is a property of a rare outcome, not a sign that the factors are weak. For two 0/1 variables, Spearman, Pearson and φ coincide, and φ can only reach 1 when both have the same prevalence. A feature with prevalence q can reach at most $\varphi_{\max} = \sqrt{p(1-q)/(q(1-p))}$ with a target of prevalence $p = 0.41\%$. COPD (5.6% of adults) can therefore reach at most 0.26, and its observed 0.11 is 43% of that maximum, with an odds ratio of 15.1 (Table 28). Ever-smoking reaches 71% of its maximum of 0.08. Correlations with this target must therefore be read together with the odds ratios and rates of §4.2–4.6, never on their own. Because no single feature is strongly associated with the target, a model will need to combine them.

Table 28. Observed and largest possible correlation (φ) between each 0/1 factor and the target, with the odds ratio (all adults with the factor known; unweighted; source: `target_correlation_bound.csv`).

Factor	Adults (cases)	Prevalence	φ observed	$\varphi_{\max}$	Share of max	Odds ratio
COPD	149,826 (618)	5.6%	0.114	0.263	43%	15.1
Heart disease	149,597 (614)	6.1%	0.045	0.253	18%	4.6
Hypertension	149,766 (619)	36.6%	0.038	0.085	45%	3.3
Diabetes	149,860 (619)	10.7%	0.022	0.186	12%	2.3
Asthma	149,864 (618)	13.9%	0.009	0.160	6%	1.4
Ever smoked	145,975 (601)	37.7%	0.059	0.083	71%	7.7
Screening-age smoker	145,664 (600)	22.2%	0.066	0.120	55%	6.5
E-cigarette (ever)	146,076 (602)	16.1%	0.002	0.147	2%	1.1

4.8 RQ3 – Regional factors

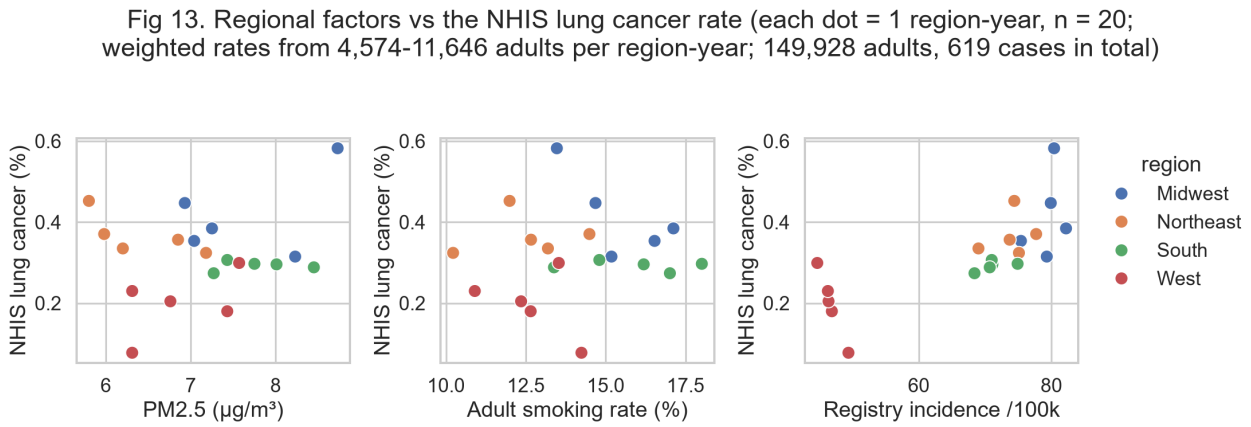


Figure 13. Regional PM2.5, adult smoking rate and registry incidence against the weighted NHIS lung-cancer rate. Each dot is one region-year ($n = 20$: 4 regions × 5 years); each region-year rate is the survey-weighted rate of its adults (4,574 to 11,646 adults per region-year; 149,928 adults and 619 cases in total).

- **The cross-source check.** Over the 20 region-years, the survey-based rate follows the official registry incidence (Fig. 13; Spearman $\rho = 0.79$, $p = 3.5 \times 10^{-5}$, Holm $p = 2.1 \times 10^{-4}$). Pooled

over five years, the weighted rate is highest in the Midwest (0.42%) and Northeast (0.37%) and lowest in the West (0.20%), the region with by far the lowest registry incidence (46.7 per 100,000 against 71–79 elsewhere).

- **Why we read it cautiously.** The 20 points are four regions measured five times, so they are not independent. Table 29 splits the correlation in two: the four regional means rank identically on both measures ($\rho = 1.0$, but $n = 4$), while the year-to-year changes within a region do not move together ($\rho = 0.03$, $p = 0.89$). The agreement is therefore mainly "the West is low on both measures". It is consistent with our self-reported target behaving like the official statistic, but it is not a formal validation, and its p-value is optimistic. The two measures also differ in kind – lifetime history vs new cases per year. For PM2.5 and the regional smoking rate, the between-region correlation in Table 29 is exactly 0.00. This is a real value, not a computation error: both measures rank the four regions differently from the NHIS rate, the squared rank differences add up to exactly 10, and $\rho = 1 - 6 \times 10/(4 \times 15) = 0$. With only four points, ρ can take just a few values $(0, \pm 0.2, \pm 0.4, \ldots, \pm 1)$, so this zero says nothing either way.
- **Air pollution and regional smoking rates show no association** (PM2.5: $\rho = -0.12$, $p = 0.63$; smoking rate: $\rho = 0.09$, $p = 0.71$), neither between nor within regions. With so few independent data points these tests are underpowered, so "no evidence" does not mean "no effect".
- **Why a null result is expected here.** The registry rate is crude, not age-adjusted (the West has a younger population); state values are averaged without population weights; and 2019–2023 pollution cannot explain cancers diagnosed many years earlier. Regional averages also say nothing about any individual (the ecological fallacy).
- **Do they add information beyond personal factors?** Mutual information on training rows (§5.5) puts all three regional columns far below the personal factors: about 0.5–0.6% of the target's information, against 10.2% for age. Their scores are above a shuffled-target noise level, but tiny. The three scores also measure the same thing. As discrete features, the regional values only say *which region-year* a person lives in: `smoking_rate` and `lung_cancer_incidence` take a different value in each of the 20 region-years, so their mutual information is exactly that of the region-year label (0.00017 nats each), and PM2.5, where two pairs of region-years share a value, is slightly lower (0.00014). So the data cannot tell whether pollution, regional smoking or registry incidence matters; at this resolution the three columns together add one small region-year signal.

Table 29. Spearman correlation of each regional measure with the weighted NHIS lung-cancer rate: over all 20 region-years, between the 4 regional means, and within regions (year-to-year deviations from each region's mean). Between-region values use $n = 4$, so ρ can only take a few values; the 0.00 for PM2.5 and smoking rate is exact (see text). Source: `regional_within_between.csv`.

Regional measure	All 20 region-years	Between 4 regions	Within regions ($n = 20$)
PM2.5	-0.12 ($p = 0.63$)	0.00	0.03 ($p = 0.89$)
Regional smoking rate	0.09 ($p = 0.71$)	0.00	-0.35 ($p = 0.13$)
Registry incidence	0.79 ($p = 3.5 \times 10^{-5}$)	1.00	0.03 ($p = 0.89$)

4.9 All tests, with multiple-testing correction

Running eighteen tests raises the chance of a false positive, so we corrected all of them together with the Holm method [8] (Table 30), which keeps the family-wise error rate at 5% and is uniformly more powerful than Bonferroni.

Table 30. All 18 hypothesis tests with raw and Holm-adjusted p-values, labelled with the research question each one answers (`reports/tables/all_tests_holm.csv`). [a]The 20 region-years are not independent (4 regions × 5 years), so this p-value is optimistic; see §4.8 and Table 29.

RQ	Test	p	Holm p	Significant
RQ1	Smoking status (χ^2)	2.2×10^{-115}	3.1×10^{-114}	yes
RQ2a	Age (Mann–Whitney U)	3.1×10^{-141}	4.6×10^{-140}	yes
RQ2b	COPD (χ^2)	$< 10^{-300}$	$< 10^{-300}$	yes
RQ2b	COPD adjusted for smoking (CMH)	1.5×10^{-208}	2.4×10^{-207}	yes
RQ2b	Heart disease (χ^2)	5.6×10^{-66}	7.3×10^{-65}	yes
RQ2b	Hypertension (χ^2)	1.3×10^{-49}	1.5×10^{-48}	yes
RQ2b	Diabetes (χ^2)	6.7×10^{-17}	6.7×10^{-16}	yes
RQ2b	Asthma (χ^2)	5.7×10^{-4}	2.9×10^{-3}	yes
RQ2b	E-cigarette, ever (χ^2)	0.39	1.00	no
RQ2c	Sex (χ^2)	0.16	0.62	no
RQ2c	Race / ethnicity (χ^2)	1.6×10^{-14}	1.1×10^{-13}	yes
RQ2c	Education (χ^2)	7.1×10^{-24}	7.8×10^{-23}	yes
RQ2c	Income vs poverty line (χ^2)	8.3×10^{-17}	7.5×10^{-16}	yes
RQ2c	BMI category (χ^2)	6.5×10^{-15}	5.2×10^{-14}	yes
RQ2c	Self-rated general health (χ^2)	1.1×10^{-248}	1.8×10^{-247}	yes
RQ3	PM2.5 vs regional rate (Spearman, $n = 20$)	0.63	1.00	no
RQ3	Smoking rate vs regional rate (Spearman, $n = 20$)	0.71	1.00	no
RQ3	Registry incidence vs regional rate (Spearman, $n = 20$)[a]	3.5×10^{-5}	2.1×10^{-4}	yes

14 of 18 tests remain significant after correction (RQ1: 1 test, RQ2a: 1, RQ2b: 7, RQ2c: 6, RQ3: 3). The four that do not – e-cigarette use, sex, PM2.5 and the regional smoking rate – are discussed above. The COPD χ^2 p-value is reported as "$< 10^{-300}$" because it is below the smallest number the computer can represent ($\chi^2 \approx 1{,}950$ with 1 degree of freedom). For the CMH test we compute the p-value from the χ^2 survival function, because the library's default returns 0 for such large statistics.

Survey-design sensitivity. Our tests treat respondents as independent draws, but NHIS is a stratified, clustered sample. To check how much this matters, `supplementary_analysis.py` reads the NHIS design variables (`PSTRAT`, `PPSU`) from the raw files and re-estimates the three key associations with survey-weighted logistic regression and confidence intervals clustered on the 3,160 sampling units (Table 31). The weighted odds ratios are somewhat larger than the unweighted ones, the interval widths change by a factor of 0.9 to 1.2, and every interval stays far from 1. The conclusions therefore do not depend on ignoring the design; a fully design-based analysis of every test remains a next step. The rates in our figures are survey-weighted.

Table 31. Survey-design sensitivity: odds ratios from unweighted logistic regression (exact; the 2 × 2 tables elsewhere add 0.5 to each cell, e.g. 7.66 instead of 7.69 for smoking) and from survey-weighted logistic regression with confidence intervals clustered on the NHIS sampling units (year × stratum × PSU). Source: `design_based_odds_ratios.csv`.

Exposure	Adults	Unweighted OR [95% CI]	Weighted, clustered OR [95% CI]
Ever smoked (vs never)	145,975	7.69 [6.24, 9.48]	9.14 [7.17, 11.65]
COPD (vs none)	149,826	15.10 [12.87, 17.71]	18.64 [15.43, 22.53]
Age, per 10 years	149,575	2.03 [1.91, 2.16]	2.15 [2.03, 2.27]

4.10 What the statistics do not show

- **No causes.** Every result here is an association in cross-sectional survey data. That smoking causes lung cancer is established elsewhere [6, 7]; our data only show a pattern consistent with

it.

- **No individual risk.** A 7.7× odds ratio describes groups. It does not say how likely any one person is to develop lung cancer, and the regional results say nothing about individuals at all.
- **No dose–response.** Without pack-years we cannot test whether more smoking means more risk, which is the core of real screening rules.
- **"Not significant" is not "no effect".** For e-cigarettes we ran a power check: with 23,470 ever-users, 122,606 never-users and a 0.41% base rate, the test had 80% power ($\alpha = 0.05$) to detect an odds ratio of about 1.34 or larger. So we can rule out a large effect in these data, but not a small one, and certainly not a long-term one that has not had time to appear.
- **Limited adjustment.** Smoking was adjusted for age, and COPD for smoking and age, by stratification. The other odds ratios are crude, so part of each one is age; a multivariable model (§7.1) is needed to separate them all at once.

5. Feature Engineering

We hand the modelling phase 38 features (Table 43), every one decided on training rows only, none of which can leak the answer, and none of which is an exact or near-exact linear combination of the others. The final tables are `nhis_lung_cancer_model_ready.csv` (original units) and `nhis_lung_cancer_model_ready_scaled.csv` (standardised), and every keep/drop decision, with its reason, is recorded in `final_features.csv`, so anyone can audit it.

5.1 Features we created

Table 32. Engineered and joined features, with their definition and rationale.

Feature	Definition	Why we built it
`screening_age_smoker`	1 if aged 50–80 and ever smoked	a proxy for the official USPSTF screening group; Fig. 8 shows that risk piles up exactly there. Without pack-years it is only an approximation (an interaction feature built from domain knowledge)
`comorbidity_count`	number of the 5 conditions (0–5)	one number for overall health burden, used in the EDA (Tables 12–13, Fig. 4, Fig. 12); not a model feature, because it is the exact sum of the five condition flags (dropped by rule 4, §5.4)
`ever_smoker`	current or former smoker	simple exposure flag, used for the odds ratio in RQ1 (dropped from the model set as redundant, §5.4)
`age_group`	18–44, 45–54, 55–64, 65–74, 75+	used for reporting, the age × smoking table and the age-adjusted tests; exact age is kept for the model instead
`pm25_mean`, `smoking_rate`, `lung_cancer_incidence`	regional values joined on region + year (§3.5)	environmental and baseline-risk context for RQ3

5.2 Encoding and imputation

- **One-hot encoding** for the nominal variables (sex, race/ethnicity, region, smoking status) and for the grouped ordinal ones (education, income, general health, BMI, age group). We chose one-hot over label encoding so the model never assumes a false order or false equal spacing between categories (for example that "Asian" is "greater" than "Black"). A complete set of dummies always adds up to 1, which makes a linear model impossible to fit, so the model-ready file drops one **reference category** per group and every other dummy is read relative to it:

race/ethnicity *White* (the largest group), education *High school / GED*, income *High* (four times the poverty line or more), BMI *Healthy*, general health *Excellent*, smoking status *Never*; for sex (*Male*) and region (*West*) the reference was already dropped by the near-duplicate rule. The choice is recorded for every dummy in `final_features.csv`.

- **Unknown answers and the reference group.** *Unknown* categories that survive selection (education, BMI) are kept as their own dummy. For smoking status, the small *Unknown* dummy was dropped as a near-duplicate of `screening_age_smoker_unknown` ($r = 0.96$), so the 2.6% of adults with an unknown smoking status now fall into the *Never* reference group and are identified only by `screening_age_smoker_unknown`, which is therefore kept (§5.7). The same happens to the 0.04% with unknown general health (reference *Excellent*).
- **Missing values** become an explicit Unknown category instead of being dropped or guessed.
- **Binary health columns** stay 0/1, with missing values set to 0 plus an `_unknown` flag; the flags for very rare gaps are later removed by the near-constant filter (§5.4).
- **Age** is filled with the training median (54 years); the `age_missing` flag created at the same time (0.24% of adults) is later removed by the near-constant filter (§5.4).

5.3 Scaling

Four numeric features use completely different units: age in years, PM2.5 in µg/m^3, smoking prevalence in %, and incidence per 100,000. (The 0–5 comorbidity count is no longer a model feature, §5.4.) Distance-based and regularised models are sensitive to these scales, so we standardise them with a `StandardScaler` fitted on training rows only. Binary columns are already on a 0–1 scale, so we leave them alone and keep them interpretable.

Table 33. Scaler parameters learned on the 119,942 training rows, and the mean of each scaled column on the test rows.

Column	Training mean	Training SD	Scaled test mean
age	52.81	18.33	−0.002
pm25_mean	7.289	0.763	0.006
smoking_rate	14.475	2.078	−0.001
lung_cancer_incidence	67.31	12.50	0.019

5.4 Feature selection

We went from 67 candidates to 38 features, applying the filter pass from the course [17, 19] on training rows only, in this order (Table 34):

The near-constant rule removes columns that describe fewer than 1 adult in 200 (for example, "COPD answer unknown" applies to 0.06% of adults). Such columns add almost no information but add dimensions – the curse of dimensionality discussed in the course. We deliberately did not use target-based filtering to drop weak features: with a 0.41% outcome, univariate relevance is unstable, and interactions (such as age × smoking) can make individually weak features useful. Relevance is instead checked in §5.5 and will be tested by ablation in the modelling phase.

Rule 4: dependencies spread over several columns (VIF). Rules 1–3 compare columns one pair at a time, so they cannot see a column that is a combination of *several* others. The variance inflation factor can: $\mathrm{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 comes from regressing feature j on all the other features (with an intercept) on the training rows. VIF = 1 means feature j is unrelated to the others, VIF > 10 is the usual alarm level, and VIF = ∞ means an exact linear dependency. After rules 1–3, 28 of the 48 features had an infinite VIF and 42 a VIF above 10 (Table 35; `reports/tables/vif_before.csv`). The causes were the comorbidity count (the sum of five flags)

Table 34. Feature-selection rules and the features each rule removed (29 in total). Never candidates: `person_id` (identifier), `split` (train/test label), `survey_weight` (a sampling weight, not a property of the person) and `CANEV_A` (reveals the answer). `sex_Male` and `region_West`, dropped by rule 3, also serve as the reference categories of sex and region.

Rule	Dropped	Features
1. Redundant by construction	8	ever_smoker, ever_smoker_unknown, age_group_18-44, age_group_45-54, age_group_55-64, age_group_65-74, age_group_75+, age_group_Unknown
2. Constant in training rows	0	–
2b. Near-constant: more than 99.5% of training rows share one value	8	age_missing, copd_unknown, asthma_unknown, hypertension_unknown, heart_disease_unknown, diabetes_unknown, sex_Unknown, general_health_Unknown
3. Near-duplicates: $\|r\| > 0.95$ with a column already kept	3	sex_Male, region_West, smoking_status_Unknown
4a. Exact linear combination of other features (VIF $= \infty$)	1	comorbidity_count
4b. Reference category of each one-hot group	6	race_ethnicity_White, education_High school / GED, income_group_High, general_health_Excellent, bmi_category_Healthy, smoking_status_Never
4c. Near-collinear block (region dummies vs regional columns)	3	region_Midwest, region_Northeast, region_South
Total dropped	29	38 features kept

and the four complete one-hot groups (race, education, income, BMI), each of which summed to 1. General health and smoking status almost did (VIF 100–643, because only their small Unknown dummies had been removed), and the region dummies almost determined the registry incidence (VIF 24–39). Rule 4 removes these dependencies (`comorbidity_count`, one reference category per group, the three region dummies). After it, the largest VIF is 7.5, nothing exceeds 10, and the PCA has no near-zero component left (§5.6; `vif_after.csv`). The two features closest to the limit are `screening_age_smoker_unknown` (7.49) and `ever_ecigarette_unknown` (7.45). They overlap because both flag the adults who skipped the smoking questions, and they are kept because the first is the only marker of an unknown smoking status. The `screening_age_smoker` flag, built from age and smoking, has a VIF of only 2.2. None of these drops removes information: an exact dependency is a direction with zero variance, and the region dummies added almost nothing beyond the 20 region-year values of the regional columns.

5.5 Relevance check with mutual information

Mutual information (MI) measures how much knowing a feature reduces uncertainty about the target, including non-linear relationships. We computed it on training rows only and treated every column with 25 or fewer distinct values as discrete; only age is treated as continuous. This matters: the continuous (nearest-neighbour) estimator is biased upwards on heavily tied values, and an earlier version of the analysis that treated the regional columns (20 distinct values each) as continuous inflated their score about tenfold (0.0016–0.0022 instead of 0.00014–0.00017), placing them wrongly in places 4–6.

With the correct estimator, the ranking agrees with the EDA: age (0.0027) and COPD (0.0026) lead, followed by the screening-age-smoker flag (0.0016), poor self-rated health, former smoking and hypertension (Fig. 14; 119,942 training rows, 495 cases). The raw values look small, but MI is bounded by the uncertainty of the target, which is tiny for a 0.41% outcome (an entropy of 0.0268 nats). Expressed as a share of that maximum, age carries 10.2% of the available information and COPD 9.8%, against 0.5–0.6% for each regional column (Table 36). To check that these scores are not noise, we shuffled the target 100 times and recomputed MI: every listed feature scores above

Table 35. Variance inflation factor of every feature after rules 1–3 (48 features) and after rule 4 (38 features), on training rows; one-hot groups are summarised by their range. ∞ = exact linear dependency.

Feature	VIF after rules 1–3	VIF after rule 4
asthma	∞	1.06
bmi_category_* (5 dummies)	∞	1.04–1.59 (Healthy dropped, 4b)
comorbidity_count	∞	dropped (4a)
copd	∞	1.18
diabetes	∞	1.17
education_* (6 dummies)	∞	1.02–1.67 (High school / GED dropped, 4b)
heart_disease	∞	1.14
hypertension	∞	1.41
income_group_* (4 dummies)	∞	1.32–1.42 (High dropped, 4b)
race_ethnicity_* (7 dummies)	∞	1.01–1.24 (White dropped, 4b)
age	1.70	1.70
ever_ecigarette	1.32	1.32
ever_ecigarette_unknown	15.86	7.45
general_health_* (5 dummies)	100.85–643.10	1.34–1.91 (Excellent dropped, 4b)
lung_cancer_incidence	33.61	1.38
pm25_mean	2.44	1.14
region_* (3 dummies)	23.98–39.02	dropped (Midwest, South, Northeast dropped, 4c)
screening_age_smoker	2.18	2.18
screening_age_smoker_unknown	13.11	7.49
sex_Female	1.07	1.07
smoking_rate	2.74	1.34
smoking_status_* (3 dummies)	113.44–258.91	1.72–2.03 (Never dropped, 4b)
Maximum (finite)	643.10	**7.49**

the 95th percentile of its shuffled scores, but the regional columns only just – their real information is a small fraction of the personal factors'. The three regional columns now occupy the last three places of the top 15 (Fig. 14). Their MI values did not change; they moved up only because ten features left the set. `smoking_rate` and `lung_cancer_incidence` have *identical* scores and null percentiles, and PM2.5 is slightly lower, for a structural reason. As discrete features, the regional columns only label the region-year a person lives in: the first two take a different value in each of the 20 region-years, so both equal the mutual information of the region-year label itself (0.000173 nats). PM2.5 has two ties (6.31 µg/m^3 in the West in 2019 and 2023; 7.43 in the South in 2022 and the West in 2021), so it merges two pairs of region-years and scores slightly less (0.000145; `regional_mi_identity.csv`). The three regional scores therefore measure one thing, not three independent signals, which is consistent with RQ3.

(Note on the earlier ranking: the tie between `screening_age_smoker` and `smoking_status_Never` at 0.00158 in the previous version of this table was a coincidence of rounding, not a duplicate. The two 2×2 tables differ – the screening flag marks 25,794 training adults with 307 cases, never-smoking 72,861 adults with 90 cases – and their MI values only differ in the eighth decimal: 0.00158140 vs 0.00158137 nats, 3×10^{-8} apart.)

Table 36. Mutual information on training rows, as a share of the target's entropy (0.0268 nats), against a permutation null (target shuffled 100 times; the smallest possible p is $1/101 \approx 0.01$). Source: `mutual_information_checked.csv`.

Feature	MI (nats)	Share of target entropy	Null, 95th percentile	Permutation p
age	0.00274	10.2%	0.00080	0.01
copd	0.00263	9.8%	0.00002	0.01
screening_age_smoker	0.00158	5.9%	0.00001	0.01
smoking_status_Former	0.00109	4.1%	0.00001	0.01
hypertension	0.00072	2.7%	0.00002	0.01
smoking_rate	0.00017	0.6%	0.00013	0.01
lung_cancer_incidence	0.00017	0.6%	0.00013	0.01
pm25_mean	0.00014	0.5%	0.00012	0.02

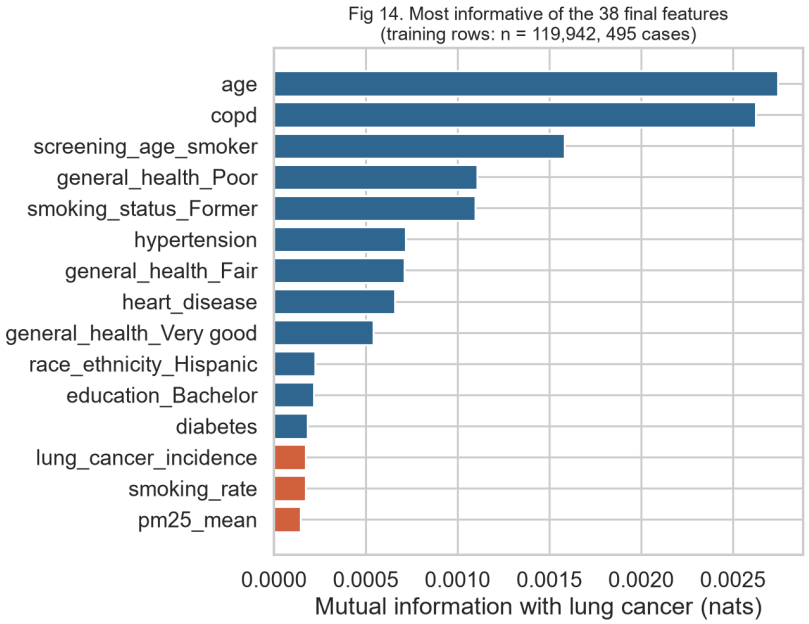


Figure 14. Mutual information between each of the 38 final features and the target on the training rows (top 15; $n = 119{,}942$ adults, 495 cases). The three regional columns (orange) take the last three places; their nearly equal values reflect one shared region-year signal (§5.5).

5.6 Feature extraction with PCA

Feature selection keeps a subset of the original columns; feature extraction builds new ones. We tested whether Principal Component Analysis could compress the 38 selected features without losing much information (Fig. 15). Because the features have different units, we standardised all of them first – equivalent to running PCA on the correlation matrix, as the course recommends [18, 19] – and fitted both the scaler and the PCA on training rows only. Loadings are correlations between a feature and a component; only loadings with an absolute value of at least 0.40, the threshold used in the course, are listed in Table 37.

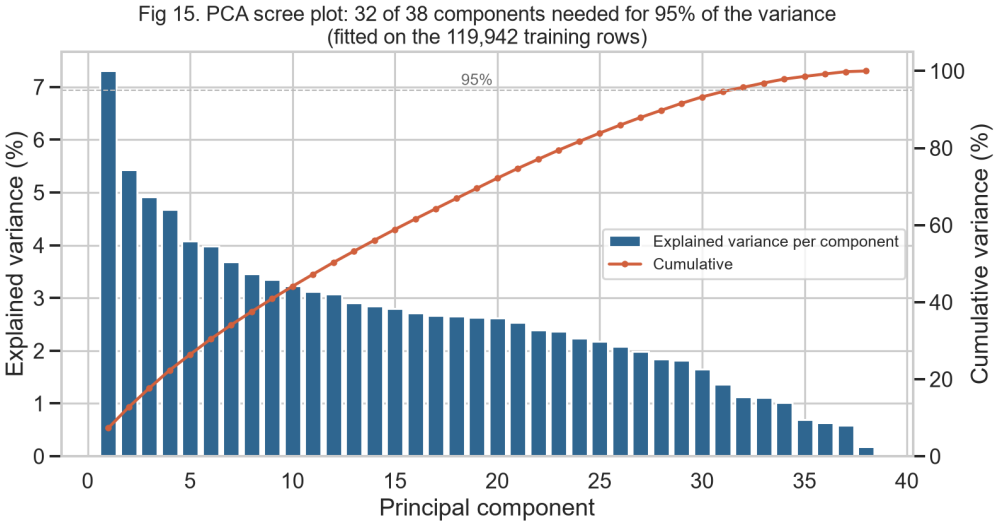


Figure 15. PCA scree plot (fitted on the 119,942 training rows): explained variance per component (bars) and cumulative variance (line). 32 of 38 components are needed to keep 95% of the variance.

Table 37. PCA summary (fitted on the 119,942 training rows).

Quantity	Value
Variance explained by PC1 / PC2 / PC3	7.3% / 5.4% / 4.9% (2.6% expected per component if all features were unrelated)
Components needed for 80% / 90% / 95% of the variance	24 / 29 / 32 of 38
Kaiser rule (eigenvalue > 1)	19 components
Components with almost zero variance (eigenvalue < 0.01)	0 (there were 8 with the previous 48 features: the dependencies removed by rule 4, §5.4)
Variance of the test rows kept by the first 32 components	95.8% (the PCA generalises; it never saw the test rows)
PC1 – "age and chronic-disease burden"	hypertension +0.59, screening_age_smoker +0.57, age +0.57, diabetes +0.46, copd +0.45, heart_disease +0.42
PC2 – "skipped the smoking questions"	screening_age_smoker_unknown +0.86, ever_ecigarette_unknown +0.86
PC3 – "older age vs current smoking and regional smoking level" (mixed)	age +0.50, smoking_rate −0.42 (next below 0.40: smoking_status_Current −0.39, ever_ecigarette −0.38, lung_cancer_incidence −0.36, pm25_mean −0.35)

Lung-cancer cases sit on the high side of PC1 (Fig. 16; mean score +2.53 vs −0.01 for other adults), which is the age-and-disease axis, but the two groups overlap heavily: two components cannot separate a rare outcome.

Decision: we keep the original features. Keeping 95% of the variance would still need 32 components – only 16% fewer columns – because the information is spread over many dimensions with no dominant one, which agrees with the weak correlations of Fig. 12c and the low VIFs of

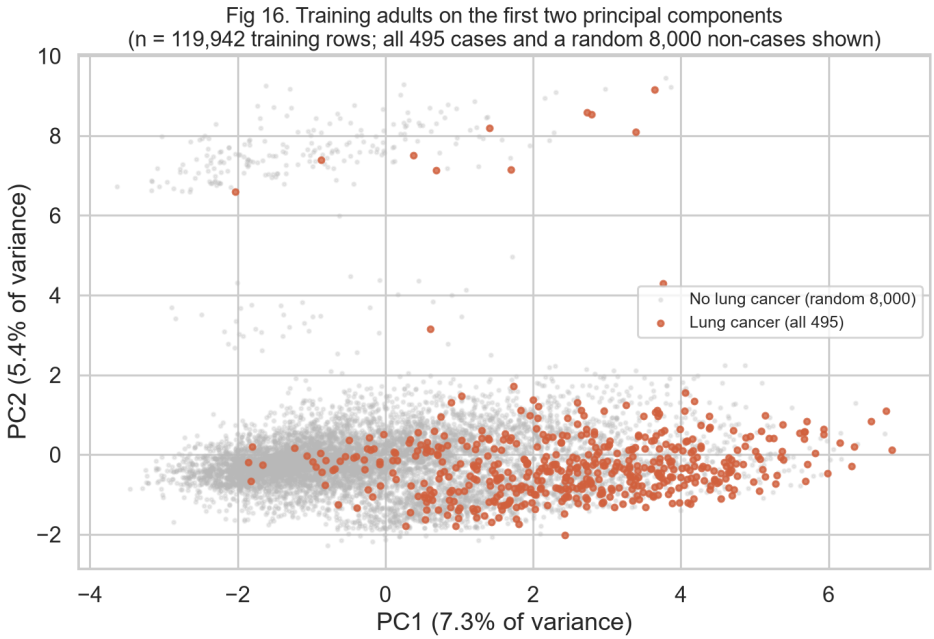


Figure 16. Training adults (n = 119,942) projected on the first two principal components: all 495 training cases (orange) and a random 8,000 non-cases (grey). The separate cloud at the top (high PC2) is the ∼2.8% of adults who skipped the smoking questions.

Table 35. Interpretability also matters for a health question: an odds ratio for "COPD" means something to a clinician, a weight on "PC4" does not. Finally, most of our columns are 0/1 indicators, for which PCA, built for continuous data, is less natural (a method for mixed data, such as factor analysis of mixed data, would be the better tool if extraction were needed). The PCA is kept as an option to compare in the modelling phase, especially for distance-based methods such as k-nearest neighbours or clustering, which suffer most from many dimensions. The previous version of the feature set had eight near-zero directions. They are gone because rule 4 removed the dependencies behind them (the comorbidity count, one reference category per one-hot group, the region dummies), and that is also why the count of components dropped: an exact dependency is a zero-variance direction that carries no information.

5.7 The final feature set and how it serves the objectives

Table 38. The 38 final features, by group (full list in Appendix B).

Group	Features	Count
Demographic	age (scaled); sex: Female (reference: Male); race/ethnicity: AIAN, AIAN and other, Asian, Black, Hispanic, Other or multiple (reference: White)	8
Socio-economic	education: Bachelor, Graduate, Less than high school, Some college, Unknown (reference: High school / GED); income vs poverty line: Below poverty, Low, Middle (reference: High)	8
Smoking and nicotine	screening_age_smoker; screening_age_smoker_unknown; ever_ecigarette; ever_ecigarette_unknown; smoking status: Current, Former (reference: Never)	6
Health	copd; asthma; hypertension; heart_disease; diabetes; general health: Fair, Good, Poor, Very good (reference: Excellent); BMI category: Obese, Overweight, Underweight, Unknown (reference: Healthy)	13
Regional context	pm25_mean (scaled); smoking_rate (scaled); lung_cancer_incidence (scaled)	3
Total		**38**

- It carries what the EDA found to matter – age, smoking, COPD and the other chronic conditions (whose sum, `comorbidity_count`, stays in the clean dataset for the EDA only) – in model-ready form, with no missing values and no exact or near-exact multicollinearity (maximum VIF 7.5).
- It links the dataset to the real clinical rule through `screening_age_smoker`, the baseline any future model must beat (Table 41).
- It is leakage-free by construction and by check: every learned step saw training rows only, so the test score of a future model will be an honest estimate.
- It keeps the regional context for an ablation test rather than discarding it on the basis of four regions.

Features we are cautious about. `general_health`, BMI, current smoking status and the "ever told" conditions may partly be consequences of a cancer diagnosis (§4.6). A model leaning on them would mostly recognise people who are already sick rather than give an early risk indication. We keep them, but plan to compare models with and without them. Race/ethnicity is kept to describe the population and to allow a fairness audit of any future model, not because race is a cause of lung cancer.

Reading the dummies. Every dummy is read against its group's reference category (Table 38). For smoking, the reference group *Never* also contains the 2.6% of adults whose smoking status is unknown. Their own dummy was dropped as a near-duplicate of `screening_age_smoker_unknown` ($r = 0.96$, rule 3), so that flag is the **only** feature that tells a model these adults are not known never-smokers. It must stay in any model that uses the smoking dummies, and an adjusted odds ratio for *Current* or *Former* smoking should be described as "vs never or unknown, holding the unknown flag fixed".

6. Discussion

6.1 Answers to the research questions

Table 39 answers the five research questions with the main numbers from Sections 2.5 and 4.

6.2 Did we meet our success criteria?

All eight criteria are met (Table 40), and all eight are verified by the code rather than by hand. The pipeline also met its reproducibility rule: on a fresh environment, `python run_all.py` rebuilt every dataset, table and figure in about 100 seconds (including the supplementary analysis), well under the five-minute target. The honest-metrics rule carries over into the modelling plan below.

These criteria check that the pipeline is correct; they do not measure how good a future model will be. For that, the modelling phase inherits a concrete benchmark: the USPSTF-style rule (Table 41).

6.3 Limitations

1. **Associations, not causes.** The data are self-reported and cross-sectional, so we cannot claim causation, and reverse causation affects smoking status, BMI, general health and the "ever told" conditions.
2. **No smoking dose.** NHIS has no pack-years or quit dates – the very variables the real screening rules use – so `screening_age_smoker` is only a proxy.
3. **Survey design only partly used.** Our main tests treat respondents as independent; a weighted, cluster-robust check of the three key odds ratios (Table 31) leaves the conclusions unchanged, but the other confidence intervals are still somewhat too narrow (the rates in the figures are weighted).

Table 39. What we found for each research question.

RQ	Answer
RQ1 – Smoking	Strong association. Ever-smokers have 7.7 times the odds of never-smokers (95% CI 6.2–9.4), and 5.8 times (4.7–7.1) after adjusting for age group; the weighted rate is 0.08% for never-smokers vs 0.81% for former and 0.60% for current smokers. Former smokers ranking above current ones reflects quitting after diagnosis (reverse causation) and age, not a protective effect of smoking.
RQ2a – Age	Median 72 vs 54 years, rank-biserial $r = 0.59$ (a random case is older than a random non-case 79.5% of the time); age and smoking act together (2.15–2.37% of ever-smokers aged 75+ vs 0.48% of never-smokers the same age).
RQ2b – Chronic conditions and e-cigarettes	COPD is the strongest association (odds ratio 15.1; among adults with known smoking status the crude 14.8 falls to 9.3 after adjusting for smoking, consistently across smoking groups, and among those whose age is also known to 5.7 after adjusting for smoking and age group); heart disease, hypertension and diabetes are moderate (2.3–4.6), partly through age; asthma is negligible; e-cigarette use shows nothing.
RQ2c – Socio-demographic and health factors	A clear education and income gradient, partly explained by smoking; the highest rates in underweight adults and those in poor health, most likely reverse causation; no difference by sex; race/ethnicity differences that largely follow age and smoking profiles.
RQ3 – Regional factors	The survey rate agrees with the official registry incidence across regions ($\rho = 0.79$ over 20 region-years, $p = 3.5 \times 10^{-5}$), but the agreement comes from the four regional averages (mainly the low West), not from year-to-year changes within regions ($\rho = 0.03$): consistent with a valid target, not proof of it. PM2.5 and regional smoking rates show no detectable association. Regional features carry far less information than personal ones (0.5–0.6% vs 10.2% of the target's information for age), and the three regional columns carry one shared region-year signal, so the data cannot separate the effect of pollution, regional smoking and registry incidence.

Table 40. Success criteria (Table 2) and the evidence that each is met (`success_criteria_check.csv`; criterion 5 is asserted in `supplementary_analysis.py`).

#	Criterion	Observed	Met
1	No duplicate person IDs	0	yes
2	No missing values in the model-ready file	0 missing cells	yes
3	Regional join adds no rows	149,928 → 149,928	yes
4	Same case rate in train and test	0.41% vs 0.41%	yes
5	No test information in fitted steps	imputation, selection (incl. the VIF step), scaling and PCA fitted on training rows; scaler asserted in code	yes
6	Every RQ tested, with effect sizes and Holm correction	18 tests: RQ1 (1), RQ2a (1), RQ2b (7), RQ2c (6), RQ3 (3)	yes
7	Every feature has a KEEP / DROP reason	67 of 67 candidates	yes
8	CANEV_A never a feature	absent	yes

4. **Rare outcome.** 619 cases limit subgroup analysis (some race groups had to be merged; the youngest age stratum has 13 cases) and will make model estimates unstable.
5. **Coarse regional data.** Regional variables are ecological, with only 20 distinct values from four regions; the registry rate is crude rather than age-adjusted and states are averaged without population weights.
6. **Lifetime target.** "Ever told" mixes recent and old diagnoses and misses people who have died, so long-term survivors are over-represented (survivorship bias), and current pollution cannot explain past diagnoses.
7. **Limited adjustment.** Smoking was adjusted for age and COPD for smoking and age, by stratification; all other associations are crude, and ordered variables were tested as nominal.
8. **Pooled years.** The 2019 questionnaire redesign and the 2020 pandemic disruption could affect comparability, although the case rate does not differ significantly between years.

6.4 Ethical considerations

NHIS is public, de-identified data, and we never attempt to re-identify anyone; household numbers are only used to check for duplicates. Because lung cancer is rare and our features are self-reported, any future model built on this work is a research prototype. It must not be presented as a screening or diagnostic tool, and false reassurance – telling someone they are at low risk – would be the most harmful error. Lung-cancer risk is also unequal across income and race groups, so a model could inherit those gaps; that is why race/ethnicity is kept for a fairness audit and why the fairness check below is part of the plan, not an optional extra.

7. Conclusion and Future Work

In US adults, a lung-cancer history concentrates in older people who have ever smoked and in people with COPD – even after accounting for smoking and age. The detailed profile of the population shows where the 619 cases are: three in four are aged 65 or older, four in five have ever smoked, and almost half have COPD. Regional pollution and smoking levels add little at the resolution available.

7.1 Next steps: the modelling phase (planned, not implemented)

1. **Baselines first.** An "always no" model and the USPSTF-style rule (`screening_age_smoker`). On the test rows the rule flags 21.9% of adults, finds 66% of the cases, and 1.25% of the people it flags have a lung-cancer history – three times the base rate (Table 41). Any real model has to beat both.
2. **Models.** Class-weighted logistic regression on the 38 features – the reference categories are already dropped and documented, and no feature has a VIF above 7.5 – for interpretable adjusted odds ratios, then tree-based models such as random forest and gradient boosting. Train on the fixed 80/20 split and tune with stratified 5-fold cross-validation on training rows only.
3. **Metrics.** Recall, precision, F1 and PR-AUC, with a threshold tuned for high recall, because missing a real case costs more than a false alarm; calibration checked on the untouched test set. No accuracy.
4. **Ablation tests.** Retrain without the reverse-causation-prone features and without the regional features, to measure what each group really contributes; compare original features with PCA components for distance-based models.
5. **Close the survey-design gap.** Extend the weighted, cluster-robust analysis (`PSTRAT`, `PPSU`) from the three key odds ratios to every association and to the model's evaluation.

6. **Use more of NHIS.** The raw files contain variables we did not use: age at lung-cancer diagnosis (`LUNGAGETC_A`, to separate recent from old diagnoses), family history of cancer and current smoking intensity.
7. **Fairness check.** Compare recall across sex, race/ethnicity and income groups before anyone relies on the model.
8. **Reporting standard.** Write up the model with the TRIPOD+AI checklist [12] and a leakage information sheet [11].

Table 41. The USPSTF-style rule (`screening_age_smoker`: aged 50–80 and ever smoked) as a benchmark: recall = share of cases flagged; precision = share of flagged adults who are cases; lift = precision divided by the base rate. Source: `screening_rule_baseline.csv`.

Rows	Adults	Flagged	Cases flagged	Recall	Precision	Lift
All adults	149,928	32,373 (21.6%)	389 of 619	62.8%	1.20%	2.9
Test rows	29,986	6,579 (21.9%)	82 of 124	66.1%	1.25%	3.0

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A. Reproducibility

We kept the project reproducible from day one, following the rule that every number and figure comes from code, never from hand edits. Code and data are in the team repository, branch main: `https://github.com/ZeinebMegaadi/Lung-cancer-early-detection`

Requirements: Python 3.11, 3.12 or 3.13 and the packages listed in `requirements.txt` (pandas, NumPy, scikit-learn, SciPy, matplotlib, seaborn, statsmodels, and jupyter, ipykernel and nbconvert for the notebook). The commands below rebuild everything:

```
git clone https://github.com/ZeinebMegaadi/Lung-cancer-early-detection.git
cd Lung-cancer-early-detection
python -m venv .venv
.venv\Scripts\Activate.ps1       (Mac / Linux: source .venv/bin/activate)
pip install -r requirements.txt
python run_all.py
python scripts/build_report.py   (this report)
```

We tested this on a fresh run: the full pipeline takes about 100 seconds, finishes without errors or warnings, and the regenerated results match the committed ones. The cleaned file `data/clean/nhis_lung_cancer_clean.csv` ($149{,}928 \times 26$) has the SHA-256 prefix `5daf9b80b848efe9` with Unix (LF) line endings, as stored in git; pandas on Windows writes Windows (CRLF) line endings, which give the prefix `2b25cb8142487dd4` for the same data. Every script can also be imported: its steps are functions, and `main()` runs them in order. The notebook `notebooks/lung_cancer_project.ipynb` uses this to walk through the six CRISP-DM sections by calling those functions, with its outputs saved. By default (`REBUILD = False`) it reuses the built files; two consecutive runs were checked to leave every output byte-identical to `run_all.py`'s. All numbers in this report come from `reports/eda_results.txt`, `reports/pca_results.txt`, `reports/supplementary_results.txt`, `reports/tables/` and `data/clean/cleaning_report.txt`; the tables that hold numbers and the key numbers in the text are generated from those files by `scripts/build_report.py`. Figure N of this report is the file `reports/figures/figNN_*.png` with the same number (Figures 12b and 12c are `fig12b_*.png` and `fig12c_*.png`).

B. Final feature list

The 38 features of `nhis_lung_cancer_model_ready.csv`, in file order. Scaled columns are standardised in `nhis_lung_cancer_model_ready_scaled.csv`; all other columns are 0/1 indicators. Reference categories (not in the file): race/ethnicity White, education High school / GED, income High, BMI Healthy, general health Excellent, smoking status Never (which also contains unknown smoking status, flagged by feature 3), sex Male; region is represented by features 11–13.

Table 42. The six pipeline steps run by `run_all.py`, the notebook, and the report build.

Step	Script	Output
0	`download_data.py`	raw NHIS, EPA, BRFSS and USCS files in `data/raw/` (skipped when already present)
1	`build_regional_data.py`	`data/clean/regional_context.csv` (20 region-years) and `regional_report.txt`
2	`build_clean_dataset.py`	clean, model-ready and scaled datasets; data dictionary; feature decisions; VIF tables (`vif_before.csv`, `vif_after.csv`); outlier table; before/after summary; cleaning report; Fig. 3
3	`eda_analysis.py`	Figures 1–2 and 4–14, including Figures 12b and 12c; test, correlation and association tables in `reports/tables/`; `reports/eda_results.txt` with every test (with its denominator) and the success-criteria check
4	`feature_extraction_pca.py`	Figures 15–16; `reports/pca_results.txt`; PCA tables
5	`supplementary_analysis.py`	dataset composition (Tables 3–5), population profile (Tables 8–13), per-year rates, age-adjusted odds ratios, regional within/between check, screening-rule baseline, mutual-information check, survey-design check and the criterion-5 assertion; `reports/supplementary_results.txt`
–	`notebooks/lung_cancer_project.ipynb`	the whole project section by section; every code cell calls a function of the scripts above (not run by `run_all.py`)
–	`build_report.py`	this report's generated tables and numbers (`report/generated/`); compiles `report/main.tex` with pdflatex (run twice) (not run by `run_all.py`)

Table 43. The 38 final features.

#	Feature	#	Feature
1	age (scaled)	20	race_ethnicity_Other or multiple
2	screening_age_smoker	21	education_Bachelor
3	screening_age_smoker_unknown	22	education_Graduate
4	ever_ecigarette	23	education_Less than high school
5	ever_ecigarette_unknown	24	education_Some college
6	copd	25	education_Unknown
7	asthma	26	income_group_Below poverty
8	hypertension	27	income_group_Low
9	heart_disease	28	income_group_Middle
10	diabetes	29	general_health_Fair
11	pm25_mean (scaled)	30	general_health_Good
12	smoking_rate (scaled)	31	general_health_Poor
13	lung_cancer_incidence (scaled)	32	general_health_Very good
14	sex_Female	33	bmi_category_Obese
15	race_ethnicity_AIAN	34	bmi_category_Overweight
16	race_ethnicity_AIAN and other	35	bmi_category_Underweight
17	race_ethnicity_Asian	36	bmi_category_Unknown
18	race_ethnicity_Black	37	smoking_status_Current
19	race_ethnicity_Hispanic	38	smoking_status_Former